

Space–Time Algebra

A compact development of paravectors and their physical applications

Abstract

This document develops a three-generator Clifford algebra and its complex paravectors, then applies them to relativity, electromagnetism, and quantum mechanics. Natural units $\hbar = c = 1$ and signature $(+, -, -, -)$ are used.

Contents

1	The algebra at a glance	3
2	Pauli-matrix representation	4
2.1	Faithful representation	4
2.2	Matrix operations and induced algebra operations	4
2.3	Algebra–matrix crosswalk	6
3	Involutions and sign conventions	7
4	Paravectors	8
4.1	Decompositions	8
4.2	Products, quadratic forms, and inverse	8
4.3	Real products and functions	10
4.4	Complex product components	10
4.5	Differentiated fields	11
5	Space–time kinematics	11
5.1	Intervals, velocity, and momentum	11
5.2	Free-particle action	13
6	Electromagnetism	13
6.1	Field equation and potentials	13
6.2	Invariants, energy, and force	15
6.3	Derivation of force from the potential and action	15
6.4	Energy balance	16

7	Relativistic wave equations	16
7.1	Paravector inner products and expectations	16
7.2	Physical spinors and minimal ideals	17
7.3	Klein–Gordon equation	18
7.4	Dirac equation in block form	18
7.5	Schrödinger–Pauli limit	18
7.6	Interacting operators and factorization	19
7.7	Low-energy derivation	20
7.8	Spin, ladder operators, and energy sectors	20
8	Rotations and Lorentz boosts	22
8.1	Rotors	22
8.2	Boosts	22
8.3	Congruences, transformed bases, and non-inertial terms	23
9	Spectral functions and quaternions	24
9.1	Functions of a unit vector	24
9.2	Quaternion subalgebra	25
10	Generalized spacetime and metric form	25
11	Reference summary	25
	Conventions and cautions	28

1 The algebra at a glance

The generators σ_n are algebraic objects with ordered multiplication rules. They are not ordinary vector components: their products generally do not commute, so the order of the factors is part of every expression. The following panel previews the generators, their product, and the general complex element developed below.

$$\begin{aligned}
\sigma_n^2 &= 1, & \sigma_n \sigma_m &= -\sigma_m \sigma_n \quad (n \neq m), \\
\mathbf{i} &= \sigma_1 \sigma_2 \sigma_3, & \mathbf{i}^2 &= -1, \\
\boldsymbol{\sigma} &= \{\sigma_1, \sigma_2, \sigma_3\}, & \mathbf{i}\boldsymbol{\sigma} &= \{\sigma_2 \sigma_3, \sigma_3 \sigma_1, \sigma_1 \sigma_2\}, \\
(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) &= \mathbf{A} \cdot \mathbf{B} + \mathbf{i}(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\
Z &= (a + b\mathbf{i}) + (\mathbf{A} + \mathbf{i}\mathbf{B}) \cdot \boldsymbol{\sigma}, & a, b \in \mathbb{R}, \quad \mathbf{A}, \mathbf{B} \in \mathbb{R}^3.
\end{aligned}$$

Ordinary three-vector operations use

$$\mathbf{A} \cdot \mathbf{B} = A_1 B_1 + A_2 B_2 + A_3 B_3, \quad (1.1)$$

$$\mathbf{A} \times \mathbf{B} = \{A_2 B_3 - A_3 B_2, A_3 B_1 - A_1 B_3, A_1 B_2 - A_2 B_1\}, \quad (1.2)$$

$$\mathbf{A}^2 := \mathbf{A} \cdot \mathbf{A}. \quad (1.3)$$

Thus the square of a vector always denotes its self dot product. The commutator is

$$[X, Y] := XY - YX, \quad [Y, X] = -[X, Y]. \quad (1.4)$$

Hence $[X, Y] = 0$ exactly when X and Y commute.

The defining relations give the graded basis

$$\underbrace{1}_{\text{scalar}}, \quad \underbrace{\sigma_1, \sigma_2, \sigma_3}_{\text{vectors}}, \quad \underbrace{\mathbf{i}\sigma_1, \mathbf{i}\sigma_2, \mathbf{i}\sigma_3}_{\text{bivectors}}, \quad \underbrace{\mathbf{i}}_{\text{pseudoscalar}}.$$

The pseudoscalar $\mathbf{i}$ commutes with every σ_n , and anticommutation gives $\mathbf{i}^2 = -\sigma_1^2 \sigma_2^2 \sigma_3^2 = -1$.

With the Kronecker delta δ_{nm} and Levi-Civita symbol ϵ_{nmk} ,

$$\sigma_n \sigma_m = \delta_{nm} + \epsilon_{nmk} \mathbf{i}\sigma_k, \quad [\sigma_n, \sigma_m] = 2\epsilon_{nmk} \mathbf{i}\sigma_k. \quad (1.5)$$

Repeated spatial indices are summed. The vector-product identity in the opening panel gives

$$[\mathbf{A} \cdot \boldsymbol{\sigma}, \mathbf{B} \cdot \boldsymbol{\sigma}] = 2\mathbf{i}(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (1.6)$$

$$\mathbf{A} \cdot \mathbf{B} = \frac{1}{2}((\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) + (\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma})). \quad (1.7)$$

The next section realizes these ordered products as ordinary 2×2 matrix products. This concrete model will define and distinguish the matrix and algebra operations used throughout the document.

2 Pauli-matrix representation

2.1 Faithful representation

Every complex element of the displayed algebra basis can be collected as

$$\begin{aligned} Z &= S + \mathbf{V} \cdot \boldsymbol{\sigma}, \\ S &\in \mathbb{C}, \quad \mathbf{V} \in \mathbb{C}^3. \end{aligned} \quad (2.1)$$

Such an element is called a complex paravector. Its faithful 2×2 complex representation begins with

$$\text{mat}(\sigma_1) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \text{mat}(\sigma_2) = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \text{mat}(\sigma_3) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Thus

$$\text{mat}(S + \mathbf{V} \cdot \boldsymbol{\sigma}) = \begin{pmatrix} S + V_3 & V_1 - iV_2 \\ V_1 + iV_2 & S - V_3 \end{pmatrix}. \quad (2.2)$$

It preserves identity, addition, and multiplied order:

$$\text{mat}(1) = \mathbf{I}_2, \quad \text{mat}(Z'Z) = \text{mat}(Z') \text{mat}(Z). \quad (2.3)$$

2.2 Matrix operations and induced algebra operations

Let $\mathbf{M} = \text{mat}(Z)$ with entries $m_{jk} = a_{jk} + b_{jk}i$, where $a_{jk}, b_{jk} \in \mathbb{R}$. Define transpose by

$$(\mathbf{M}^\top)_{jk} := m_{kj}. \quad (2.4)$$

Define entrywise complex conjugation independently by

$$m_{jk}^c := a_{jk} - b_{jk}i, \quad \mathbf{M}^c := \begin{pmatrix} m_{11}^c & m_{12}^c \\ m_{21}^c & m_{22}^c \end{pmatrix}. \quad (2.5)$$

The Hermitian transpose is then

$$\mathbf{M}^H := (\mathbf{M}^c)^\top = (\mathbf{M}^\top)^c. \quad (2.6)$$

Define the algebra operation H through the faithful representation by

$$\text{mat}(Z^H) := \mathbf{M}^H. \quad (2.7)$$

Its action on every grade is

$$\text{mat}(1)^H = \mathbf{I}_2, \quad (2.8)$$

$$(i\mathbf{I}_2)^H = -i\mathbf{I}_2, \quad (2.9)$$

$$\text{mat}(\sigma_n)^H = \text{mat}(\sigma_n), \quad n = 1, 2, 3, \quad (2.10)$$

$$\text{mat}(i\sigma_n)^H = -\text{mat}(i\sigma_n), \quad n = 1, 2, 3. \quad (2.11)$$

Hence $1^H = 1$, $i^H = -i$, $\sigma_n^H = \sigma_n$, and $(i\sigma_n)^H = -i\sigma_n$. Hermitian conjugation reverses matrix-product order. Therefore faithfulness gives

$$\text{mat}((\sigma_n \sigma_m)^H) = (\text{mat}(\sigma_n) \text{mat}(\sigma_m))^H \quad (2.12)$$

$$= \text{mat}(\sigma_m)^H \text{mat}(\sigma_n)^H = \text{mat}(\sigma_m \sigma_n), \quad (2.13)$$

and hence

$$(\sigma_n \sigma_m)^H = \sigma_m^H \sigma_n^H = \sigma_m \sigma_n. \quad (2.14)$$

Thus H is precisely reversion of multiplied sigma order. To define the distinct sigma conjugation, let $\mathbf{J} := \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and $\mathbf{J}^2 = -\mathbf{I}_2$, and set

$$\mathbf{M}^* := -\mathbf{J} \mathbf{M}^C \mathbf{J}, \quad \text{mat}(\mathbf{Z}^*) := \mathbf{M}^*. \quad (2.15)$$

This gives the required action on every grade:

$$\text{mat}(1)^* = -\mathbf{J} \mathbf{I}_2^C \mathbf{J} = \mathbf{I}_2, \quad (2.16)$$

$$(\mathbf{i} \mathbf{I}_2)^* = -\mathbf{J} (\mathbf{i} \mathbf{I}_2)^C \mathbf{J} = -\mathbf{i} \mathbf{I}_2, \quad (2.17)$$

$$\text{mat}(\sigma_n)^* = -\mathbf{J} \text{mat}(\sigma_n)^C \mathbf{J} = -\text{mat}(\sigma_n), \quad n = 1, 2, 3, \quad (2.18)$$

$$\text{mat}(\mathbf{i} \sigma_n)^* = -\mathbf{J} \text{mat}(\mathbf{i} \sigma_n)^C \mathbf{J} = \text{mat}(\mathbf{i} \sigma_n), \quad n = 1, 2, 3. \quad (2.19)$$

The operation also preserves multiplied order. Since entrywise conjugation preserves order and $\mathbf{J}^2 = -\mathbf{I}_2$,

$$(\mathbf{M} \mathbf{N})^* := -\mathbf{J} (\mathbf{M} \mathbf{N})^C \mathbf{J} = -\mathbf{J} \mathbf{M}^C \mathbf{N}^C \mathbf{J}, \quad (2.20)$$

$$\mathbf{M}^* \mathbf{N}^* = (-\mathbf{J} \mathbf{M}^C \mathbf{J}) (-\mathbf{J} \mathbf{N}^C \mathbf{J}) = -\mathbf{J} \mathbf{M}^C \mathbf{N}^C \mathbf{J}. \quad (2.21)$$

Therefore

$$(\mathbf{M} \mathbf{N})^* = \mathbf{M}^* \mathbf{N}^*. \quad (2.22)$$

The four operations must remain distinct:

symbol	matrix operation	relation to the algebra
$\mathbf{M}^T$	transpose	reverses matrix-product order
$\mathbf{M}^C$	entrywise complex conjugation	sends $\mathbf{i}$ to $-\mathbf{i}$
$\mathbf{M}^H$	Hermitian transpose	$\text{mat}(\mathbf{Z}^H)$
$\mathbf{M}^*$	sigma conjugation	$\text{mat}(\mathbf{Z}^*)$

Their product-order rules are

$$(\mathbf{M} \mathbf{N})^T = \mathbf{N}^T \mathbf{M}^T, \quad (\mathbf{M} \mathbf{N})^C = \mathbf{M}^C \mathbf{N}^C, \quad (2.23)$$

$$(\mathbf{M} \mathbf{N})^H = \mathbf{N}^H \mathbf{M}^H. \quad (2.24)$$

Sigma conjugation preserves order by equation (2.22). Thus the induced algebra operations satisfy

$$(\mathbf{Z}^H)^H = \mathbf{Z}, \quad (\mathbf{Z} \mathbf{Z}')^H = \mathbf{Z}'^H \mathbf{Z}^H, \quad (2.25)$$

$$(\mathbf{Z}^*)^* = \mathbf{Z}, \quad (\mathbf{Z} \mathbf{Z}')^* = \mathbf{Z}^* \mathbf{Z}'^*, \quad (2.26)$$

$$\mathbf{Z}^{*H} = \mathbf{Z}^{H*}. \quad (2.27)$$

Trace, determinant, and adjugate. For $\mathbf{M} = \text{mat}(Z)$, define the matrix trace and determinant by

$$\text{tr}(\mathbf{M}) := m_{11} + m_{22}, \quad \det(\mathbf{M}) := m_{11}m_{22} - m_{12}m_{21}. \quad (2.28)$$

The corresponding paravector operations are

$$\text{tr}(Z) := \text{tr}(\mathbf{M}) = 2S, \quad \det(Z) := \det(\mathbf{M}). \quad (2.29)$$

Faithfulness transfers the matrix identities to the algebra:

$$\text{tr}(Z + Z') = \text{tr}(Z) + \text{tr}(Z'), \quad \text{tr}(cZ) = c \text{tr}(Z), \quad (2.30)$$

$$\text{tr}(ZZ') = \text{tr}(Z'Z), \quad (2.31)$$

$$\det(ZZ') = \det(Z) \det(Z'), \quad \det(cZ) = c^2 \det(Z) \quad (c \in \mathbb{C}). \quad (2.32)$$

For a 2×2 matrix and its paravector preimage, define

$$\text{adj}(\mathbf{M}) := \text{tr}(\mathbf{M})\mathbf{I}_2 - \mathbf{M}, \quad (2.33)$$

$$\text{mat}(\text{adj}(Z)) := \text{adj}(\mathbf{M}). \quad (2.34)$$

Direct comparison with the induced involutions gives

$$\text{adj}(Z) = Z^{*\text{H}}, \quad \text{mat}(\text{adj}(Z)) = \text{adj}(\mathbf{M}) = (\mathbf{M}^*)^{\text{H}}. \quad (2.35)$$

Equivalently,

$$\text{adj}(Z) = \text{tr}(Z) - Z, \quad \text{tr}(Z) = Z + \text{adj}(Z) = 2S. \quad (2.36)$$

The adjugate reverses product order and obeys

$$\text{adj}(ZZ') = \text{adj}(Z') \text{adj}(Z), \quad \text{adj}(\text{adj}(Z)) = Z, \quad (2.37)$$

$$\text{adj}(Z)Z = Z \text{adj}(Z) = \det(Z), \quad \text{mat}(\text{adj}(Z)Z) = \det(\mathbf{M})\mathbf{I}_2. \quad (2.38)$$

Consequently, when $\det(Z) \neq 0$,

$$Z^{-1} = \frac{\text{adj}(Z)}{\det(Z)}. \quad (2.39)$$

Finally, Cayley–Hamilton gives

$$\mathbf{M}^2 - \text{tr}(\mathbf{M})\mathbf{M} + \det(\mathbf{M})\mathbf{I}_2 = 0, \quad (2.40)$$

$$Z^2 - \text{tr}(Z)Z + \det(Z) = 0. \quad (2.41)$$

2.3 Algebra–matrix crosswalk

For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, define

$$\text{sc}(Z) := S, \quad \text{vec}(Z) := \mathbf{V}. \quad (2.42)$$

Thus $\text{vec}(Z)$ is the coefficient vector, while its algebra element is $\mathbf{V} \cdot \boldsymbol{\sigma}$.

For $\mathbf{M} = \text{mat}(Z)$, the extraction formulas are

$$\text{sc}(Z) = S = \frac{1}{2} \text{tr}(\mathbf{M}), \quad \text{mat}(\text{sc}(Z)) = S\mathbf{I}_2, \quad (2.43)$$

$$(\text{vec}(Z))_n = V_n = \frac{1}{2} \text{tr}(\boldsymbol{\sigma}_n \mathbf{M}), \quad \text{mat}(\text{vec}(Z) \cdot \boldsymbol{\sigma}) = \mathbf{M} - \frac{1}{2} \text{tr}(\mathbf{M})\mathbf{I}_2. \quad (2.44)$$

Define real and imaginary paravector extraction through Hermitian splitting:

$$\text{mat}(\text{re}(Z)) := \frac{1}{2}(\mathbf{M} + \mathbf{M}^{\text{H}}), \quad (2.45)$$

$$\text{mat}(\text{im}(Z)) := \frac{\mathbf{M} - \mathbf{M}^{\text{H}}}{2\text{i}}. \quad (2.46)$$

By construction, the algebra and matrix involutions agree through

$$\text{mat}(Z^{\text{H}}) = \mathbf{M}^{\text{H}}, \quad \text{mat}(Z^*) = \mathbf{M}^*, \quad \text{mat}(Z^{*\text{H}}) = (\mathbf{M}^*)^{\text{H}}. \quad (2.47)$$

The matrix-only operations $^{\text{C}}$ and $^{\text{T}}$ are related to the algebra operations by

$$\mathbf{M}^{\text{C}} = -\mathbf{J}\mathbf{M}^* \mathbf{J}, \quad \mathbf{M}^{\text{T}} = -\mathbf{J} \text{mat}(Z^{*\text{H}}) \mathbf{J}. \quad (2.48)$$

Finally, scalar extraction of a product and the paravector coefficient norm are

$$\text{sc}(Z'Z) = \frac{1}{2} \text{tr}(\mathbf{M}_{Z'} \mathbf{M}_Z), \quad (2.49)$$

$$\|Z\|^2 := \frac{1}{2} \text{tr}(\mathbf{M}^{\text{H}} \mathbf{M}) = \frac{1}{2} \text{tr}(\mathbf{M} \mathbf{M}^{\text{H}}). \quad (2.50)$$

3 Involutions and sign conventions

The matrix construction now has a direct algebraic interpretation. Sigma conjugation $(\cdot)^*$ changes the sign of every generator without reversing order. Reversion $(\cdot)^{\text{H}}$ reverses multiplied sigma order and, as shown in Section 2, becomes Hermitian conjugation. Their action by grade is

grade	0	1	2	3
representative	1	σ_n	$\text{i}\sigma_n$	i
$(\cdot)^*$	+	-	+	-
$(\cdot)^{\text{H}}$	+	+	-	-
$(\cdot)^{*\text{H}}$	+	-	-	+

In particular,

$$\sigma_n^* = -\sigma_n, \quad \sigma_n^{\text{H}} = \sigma_n, \quad \text{i}^* = -\text{i}, \quad \text{i}^{\text{H}} = -\text{i}, \quad (3.1)$$

$$(\text{i}\sigma_n)^* = \text{i}^* \sigma_n^* = \text{i}\sigma_n, \quad (\text{i}\sigma_n)^{\text{H}} = \sigma_n^{\text{H}} \text{i}^{\text{H}} = -\text{i}\sigma_n. \quad (3.2)$$

The pseudoscalar is central: moving σ_n through the other two generators produces two sign changes, so

$$\text{i}\sigma_n = \sigma_n \text{i}. \quad (3.3)$$

Sigma conjugation preserves order, whereas reversion reverses it. Hence

$$(\sigma_m \sigma_n)^* = \sigma_m^* \sigma_n^* = (-\sigma_m)(-\sigma_n) = \sigma_m \sigma_n, \quad (3.4)$$

$$(\sigma_m \sigma_n)^{\text{H}} = \sigma_n^{\text{H}} \sigma_m^{\text{H}} = \sigma_n \sigma_m. \quad (3.5)$$

They obey

$$(ZZ')^{\text{H}} = Z'^{\text{H}} Z^{\text{H}}, \quad (ZZ')^* = Z^* Z'^*, \quad Z^{*\text{H}} = Z^{\text{H}*}.$$

Thus the adjugate $\text{adj}(Z) = Z^{*\text{H}}$ introduced in Section 2 changes the sign of the vector part of a paravector.

4 Paravectors

4.1 Decompositions

Resolve the general complex paravector into real coefficients:

$$Z = (a + bi) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma} = X + iY = S + \mathbf{V} \cdot \boldsymbol{\sigma}, \quad (4.1)$$

where $a, b \in \mathbb{R}$, $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$, $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$, $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$, $S = a + bi$, and $\mathbf{V} = \mathbf{A} + i\mathbf{B}$. The involutions give

$$Z^* = (a - bi) + (-\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (4.2)$$

$$Z^H = (a - bi) + (\mathbf{A} - i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (4.3)$$

$$Z^{*H} = (a + bi) - (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (4.4)$$

The extraction functions defined in Section 2 give

$$\text{sc}(Z) = S \quad (\text{scalar part}), \quad \text{vec}(Z) = \mathbf{V} \quad (\text{vector coefficient}), \quad (4.5)$$

$$\text{re}(Z) = X \quad (\text{real paravector part}), \quad \text{im}(Z) = Y \quad (\text{imaginary paravector part}). \quad (4.6)$$

The functions re and im also act componentwise on complex scalars and vectors. Thus the components are recovered by

$$S = \text{sc}(Z) = \frac{1}{2}(Z + Z^{*H}), \quad \mathbf{V} = \text{vec}(Z), \quad \mathbf{V} \cdot \boldsymbol{\sigma} = \frac{1}{2}(Z - Z^{*H}), \quad (4.7)$$

$$X = \text{re}(Z) = \frac{1}{2}(Z + Z^H), \quad Y = \text{im}(Z) = \frac{Z - Z^H}{2i}. \quad (4.8)$$

The four real coefficient groups then follow by composition:

$$a = \text{re}(S) = \text{sc}(X) = \text{sc}(\text{re}(Z)), \quad b = \text{im}(S) = \text{sc}(Y) = \text{sc}(\text{im}(Z)), \quad (4.9)$$

$$\mathbf{A} = \text{re}(\mathbf{V}) = \text{vec}(X) = \text{vec}(\text{re}(Z)), \quad \mathbf{B} = \text{im}(\mathbf{V}) = \text{vec}(Y) = \text{vec}(\text{im}(Z)). \quad (4.10)$$

Equivalently, they may be recovered directly from the involutions:

$$a = \frac{1}{4}(Z + Z^* + Z^H + Z^{*H}), \quad (4.11)$$

$$b = \frac{1}{4i}(Z - Z^* - Z^H + Z^{*H}), \quad (4.12)$$

$$\mathbf{A} \cdot \boldsymbol{\sigma} = \frac{1}{4}(Z - Z^* + Z^H - Z^{*H}), \quad (4.13)$$

$$\mathbf{B} \cdot \boldsymbol{\sigma} = \frac{1}{4i}(Z + Z^* - Z^H - Z^{*H}). \quad (4.14)$$

4.2 Products, quadratic forms, and inverse

For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ and $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$,

$$ZZ' = (SS' + \mathbf{V} \cdot \mathbf{V}') + (S\mathbf{V}' + S'\mathbf{V} + i\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma}. \quad (4.15)$$

Thus

$$\text{sc}(ZZ') = SS' + \mathbf{V} \cdot \mathbf{V}', \quad (4.16)$$

$$\text{vec}(ZZ') = S\mathbf{V}' + S'\mathbf{V} + i\mathbf{V} \times \mathbf{V}'. \quad (4.17)$$

Equivalently, if $Z = X + iY$ and $Z' = X' + iY'$, then

$$ZZ' = (XX' - YY') + i(XY' + YX').$$

The determinant is

$$\det(Z) = \text{adj}(Z) Z = Z \text{adj}(Z) = S^2 - \mathbf{V} \cdot \mathbf{V}. \quad (4.18)$$

Define the scalar, vector, and paravector coefficient norms by

$$\|S\|^2 := SS^* = S^*S, \quad (4.19)$$

$$\|\mathbf{V}\|^2 := \mathbf{V} \cdot \mathbf{V}^* = \mathbf{V}^* \cdot \mathbf{V}, \quad (4.20)$$

$$\|Z\|^2 = \|S\|^2 + \|\mathbf{V}\|^2. \quad (4.21)$$

Equivalently,

$$\|Z\|^2 = \text{sc}(Z^H Z) = \text{sc}(Z Z^H) = \text{sc}(Z^* Z^{*H}) = \text{sc}(Z^{*H} Z^*). \quad (4.22)$$

Thus $\det(Z)$ may be complex, whereas $\|Z\|^2$ is real and nonnegative. For $S = a + ib$ and $\mathbf{V} = \mathbf{A} + i\mathbf{B}$,

$$\|S\|^2 = a^2 + b^2, \quad \|\mathbf{V}\|^2 = \mathbf{A}^2 + \mathbf{B}^2. \quad (4.23)$$

In general,

$$S^2 \neq \|S\|^2, \quad \mathbf{V}^2 \neq \|\mathbf{V}\|^2. \quad (4.24)$$

Whenever $\det(Z) \neq 0$,

$$Z^{-1} = \frac{\text{adj}(Z)}{\det(Z)}. \quad (4.25)$$

For any complex scalar c ,

$$\det(cZ) = c^2 \det(Z). \quad (4.26)$$

Indeed, $\text{mat}(cZ) = c \text{mat}(Z)$, and the determinant of a 2×2 matrix is quadratic in a scalar multiplier. Thus, for real r and real phase θ ,

$$\det(re^{i\theta} Z) = r^2 e^{2i\theta} \det(Z), \quad (4.27)$$

$$\|re^{-i\theta} Z\|^2 = r^2 \|Z\|^2. \quad (4.28)$$

Every paravector satisfies the quadratic identity

$$Z^2 - 2SZ + (S^2 - \mathbf{V}^2) = 0. \quad (4.29)$$

Its two algebraic eigenvalues are $\lambda_{\pm} = S \pm \sqrt{\mathbf{V}^2}$.

Vector-product derivation. Using $\sigma_n \sigma_m = \delta_{nm} + \epsilon_{nmk} i \sigma_k$,

$$(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) = \sum_{n,m} A_n B_m \sigma_n \sigma_m \quad (4.30)$$

$$= \sum_{n,m} A_n B_m (\delta_{nm} + \epsilon_{nmk} i \sigma_k) \quad (4.31)$$

$$= \mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (4.32)$$

Hence

$$(\mathbf{A} \cdot \boldsymbol{\sigma})^2 = \mathbf{A}^2, \quad (\mathbf{A} \cdot \boldsymbol{\sigma})^{2n} = (\mathbf{A}^2)^n, \quad (\mathbf{A} \cdot \boldsymbol{\sigma})^{2n+1} = (\mathbf{A}^2)^n \mathbf{A} \cdot \boldsymbol{\sigma}, \quad (4.33)$$

$$(\mathbf{A} \cdot \boldsymbol{\sigma}) \boldsymbol{\sigma} = \mathbf{A} + i \mathbf{A} \times \boldsymbol{\sigma}, \quad [\mathbf{A} \cdot \boldsymbol{\sigma}, \boldsymbol{\sigma}] = -2i \mathbf{A} \times \boldsymbol{\sigma}, \quad (4.34)$$

$$[\mathbf{A} \cdot i \boldsymbol{\sigma}, \boldsymbol{\sigma}] = 2 \mathbf{A} \times \boldsymbol{\sigma}, \quad [\mathbf{A} \cdot \boldsymbol{\sigma}, \boldsymbol{\sigma}] \cdot \mathbf{B} = 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (4.35)$$

4.3 Real products and functions

For $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$ and $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$,

$$XY = (ab + \mathbf{A} \cdot \mathbf{B}) + (a\mathbf{B} + b\mathbf{A} + \mathbf{i}\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (4.36)$$

$$\text{sc}(XY) = ab + \mathbf{A} \cdot \mathbf{B}, \quad (4.37)$$

$$\text{vec}(XY) = a\mathbf{B} + b\mathbf{A} + \mathbf{i}\mathbf{A} \times \mathbf{B}, \quad (4.38)$$

$$[X, Y] = 2\mathbf{i}(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (4.39)$$

$$X^2 = (a^2 + \mathbf{A}^2) + 2a\mathbf{A} \cdot \boldsymbol{\sigma}, \quad (4.40)$$

$$\text{sc}(X^2) = a^2 + \mathbf{A}^2, \quad (4.41)$$

$$\text{vec}(X^2) = 2a\mathbf{A}, \quad (4.42)$$

$$X^2 - 2aX + (a^2 - \mathbf{A}^2) = 0, \quad (4.43)$$

$$X^{-1} = \frac{a - \mathbf{A} \cdot \boldsymbol{\sigma}}{a^2 - \mathbf{A}^2}, \quad (4.44)$$

$$\text{d}(X^{-1}) = -X^{-1}(\text{d}X)X^{-1}. \quad (4.45)$$

The condition $\mathbf{A} \times \mathbf{B} = 0$ implies $XY = YX$, $X^{-1}Y = YX^{-1}$, and $\det(XY) = \det(Y)\det(X)$. If $X^2 = Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$, then

$$a^2 = \frac{1}{2} \left(b \pm \sqrt{b^2 - \mathbf{B}^2} \right), \quad \mathbf{A}^2 = b - a^2,$$

with signs and vector direction chosen consistently. Conversely, in the real domain where $b^2 > \mathbf{B}^2$, $b > 0$, and $\mathbf{B} \neq 0$,

$$\log Y = \frac{1}{2} \log(b^2 - \mathbf{B}^2) + \text{atanh}\left(\frac{\|\mathbf{B}\|}{b}\right) \frac{\mathbf{B}}{\|\mathbf{B}\|} \cdot \boldsymbol{\sigma}. \quad (4.46)$$

Exponentials. For $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$ and $\mathbf{u} = \mathbf{A}/\|\mathbf{A}\|$,

$$e^X = e^a (\cosh\|\mathbf{A}\| + \sinh\|\mathbf{A}\| \mathbf{u} \cdot \boldsymbol{\sigma}), \quad (4.47)$$

$$e^{-\mathbf{i}\mathbf{A} \cdot \boldsymbol{\sigma}} = \cos\|\mathbf{A}\| - \mathbf{i} \sin\|\mathbf{A}\| \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (4.48)$$

For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ and $z = \sqrt{\mathbf{V}^2}$,

$$e^Z = e^S \left(\cosh z + \frac{\sinh z}{z} \mathbf{V} \cdot \boldsymbol{\sigma} \right), \quad (4.49)$$

with the quotient defined by continuity at $z = 0$.

4.4 Complex product components

Write $Z = (a + bi) + (\mathbf{A} + \mathbf{i}\mathbf{B}) \cdot \boldsymbol{\sigma}$ and use primes for a second paravector. The four real components of ZZ' are

$$\text{real scalar: } aa' - bb' + \mathbf{A} \cdot \mathbf{A}' - \mathbf{B} \cdot \mathbf{B}', \quad (4.50)$$

$$\text{imaginary scalar: } ab' + ba' + \mathbf{A} \cdot \mathbf{B}' + \mathbf{B} \cdot \mathbf{A}', \quad (4.51)$$

$$\text{real vector: } a\mathbf{A}' - b\mathbf{B}' + a'\mathbf{A} - b'\mathbf{B} - \mathbf{A} \times \mathbf{B}' - \mathbf{B} \times \mathbf{A}', \quad (4.52)$$

$$\text{imaginary vector: } a\mathbf{B}' + b\mathbf{A}' + a'\mathbf{B} + b'\mathbf{A} + \mathbf{A} \times \mathbf{A}' - \mathbf{B} \times \mathbf{B}'. \quad (4.53)$$

For the important quadratic products:

$$Z^2 : (a^2 - b^2 + \mathbf{A}^2 - \mathbf{B}^2) + 2i(ab + \mathbf{A} \cdot \mathbf{B}) + 2(a\mathbf{A} - b\mathbf{B}) \cdot \boldsymbol{\sigma} + 2i(a\mathbf{B} + b\mathbf{A}) \cdot \boldsymbol{\sigma}, \quad (4.54)$$

$$ZZ^*, Z^*Z : (a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2) \mp 2(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma} + 2i(a\mathbf{B} - b\mathbf{A}) \cdot \boldsymbol{\sigma}, \quad (4.55)$$

$$ZZ^H, Z^H Z : a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2 + 2(a\mathbf{A} + b\mathbf{B} \pm \mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (4.56)$$

$$Z \operatorname{adj}(Z) : a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2 + 2i(ab - \mathbf{A} \cdot \mathbf{B}). \quad (4.57)$$

The scalar part is cyclic, $\operatorname{sc}(ZZ') = \operatorname{sc}(Z'Z)$, and

$$(ZZ')^* = Z^*Z'^*, \quad (ZZ')^H = Z'^H Z^H, \quad (ZZ')^{*H} = Z'^{*H} Z^{*H}. \quad (4.58)$$

The paired signs follow the displayed product order. Thus $ZZ^* = Z^*Z$ and $ZZ^H = Z^H Z$ precisely when $\mathbf{A} \times \mathbf{B} = 0$.

4.5 Differentiated fields

For products of fields,

$$\partial(ZZ') = (\partial Z)Z' + Z(\partial Z') + [\boldsymbol{\sigma}, Z] \cdot \partial_{\mathbf{r}} Z'. \quad (4.59)$$

Let $F = (a + bi) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ depend on $(t, \mathbf{r})$. Then

$$\partial F = (\partial_t a + \partial_{\mathbf{r}} \cdot \mathbf{A}) + i(\partial_t b + \partial_{\mathbf{r}} \cdot \mathbf{B}) \quad (4.60)$$

$$+ (\partial_{\mathbf{r}} a + \partial_t \mathbf{A} - \partial_{\mathbf{r}} \times \mathbf{B}) \cdot \boldsymbol{\sigma} \quad (4.61)$$

$$+ i(\partial_{\mathbf{r}} b + \partial_{\mathbf{r}} \times \mathbf{A} + \partial_t \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (4.62)$$

Multiplication by i rotates these four components: the real scalar becomes $-\partial_t b - \partial_{\mathbf{r}} \cdot \mathbf{B}$, the imaginary scalar becomes $\partial_t a + \partial_{\mathbf{r}} \cdot \mathbf{A}$, the real vector becomes the negative of the imaginary vector above, and the imaginary vector becomes the real vector.

5 Space–time kinematics

5.1 Intervals, velocity, and momentum

Represent a differential event by the real paravector

$$dX = dt + d\mathbf{r} \cdot \boldsymbol{\sigma}, \quad dX^* = dt - d\mathbf{r} \cdot \boldsymbol{\sigma}. \quad (5.1)$$

The Minkowski interval is

$$ds^2 = \det(dX) = dX \, dX^* = dt^2 - d\mathbf{r}^2. \quad (5.2)$$

It is time-like, null, or space-like according as this expression is positive, zero, or negative. For a future-directed time-like interval, choose the positive scalar root

$$ds := \sqrt{\det(dX)}. \quad (5.3)$$

Thus the following kinematics explores the determinant of dX and its normalization by this scalar. For two intervals,

$$\det(dX + dX') - \det(dX) - \det(dX') = 2(dt \, dt' - d\mathbf{r} \cdot d\mathbf{r}'), \quad [dX, dX'] = 2i(d\mathbf{r} \times d\mathbf{r}') \cdot \boldsymbol{\sigma}. \quad (5.4)$$

When $ds^2 \neq 0$,

$$dX^{-1} = \frac{\text{adj}(dX)}{\det(dX)} = \frac{dX^*}{ds^2}. \quad (5.5)$$

A null interval has $dt^2 = d\mathbf{r}^2$ and no inverse. Along a time-like worldline, s is proper time. Since $d\mathbf{r} = \mathbf{v} dt$,

$$ds^2 = (1 - \mathbf{v}^2) dt^2, \quad ds = \frac{dt}{\gamma}, \quad \gamma := \frac{1}{\sqrt{1 - \mathbf{v}^2}}. \quad (5.6)$$

Hence

$$\mathbf{v} = \frac{d\mathbf{r}}{dt}, \quad \gamma = \frac{dt}{ds}, \quad U := \frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (5.7)$$

Using the scalar-scaling rule (4.26), its determinant is

$$\det(U) = \det\left(\frac{dX}{ds}\right) = \frac{\det(dX)}{ds^2} = 1. \quad (5.8)$$

Equivalently, $\det(U) = \gamma^2(1 - \mathbf{v}^2) = 1$, so

$$U^{-1} = \frac{\text{adj}(U)}{\det(U)} = U^*. \quad (5.9)$$

Proper time therefore obeys

$$\tau = \int ds = \int \sqrt{1 - \mathbf{v}^2} dt = \int \frac{dt}{\gamma}. \quad (5.10)$$

The four-momentum is the paravector

$$mU = m\gamma + m\gamma\mathbf{v} \cdot \boldsymbol{\sigma} = E + \mathbf{P} \cdot \boldsymbol{\sigma}. \quad (5.11)$$

Its scalar and vector components define

$$E := m\gamma, \quad \mathbf{P} := m\gamma\mathbf{v}. \quad (5.12)$$

Taking its determinant gives the mass-shell relation

$$\det(mU) = m^2 \det(U) = m^2 = \det(E + \mathbf{P} \cdot \boldsymbol{\sigma}) = E^2 - \mathbf{P}^2, \quad (5.13)$$

or $E^2 = \mathbf{P}^2 + m^2$; for $m = 0$, $E^2 = \mathbf{P}^2$.

For coordinate acceleration $\mathbf{a} = d\mathbf{v}/dt$,

$$\frac{d\gamma}{dt} = \gamma^3 \mathbf{v} \cdot \mathbf{a}, \quad \frac{d}{ds} = \gamma \frac{d}{dt}, \quad (5.14)$$

so

$$\frac{dU}{ds} = \gamma^4(\mathbf{v} \cdot \mathbf{a}) + (\gamma^4(\mathbf{v} \cdot \mathbf{a})\mathbf{v} + \gamma^2\mathbf{a}) \cdot \boldsymbol{\sigma}, \quad (5.15)$$

Differentiating $U^*U = 1$ gives

$$\frac{dU^*}{ds}U + U^*\frac{dU}{ds} = 0, \quad \text{sc}(U^* dU/ds) = 0. \quad (5.16)$$

If $A = dU/ds$, then

$$\det(A) = -\gamma^4(\mathbf{a}^2 + \gamma^2(\mathbf{v} \cdot \mathbf{a})^2). \quad (5.17)$$

For $\mathbf{u} = \mathbf{v}/\|\mathbf{v}\|$, define $\mathbf{a}_{\parallel} = (\mathbf{a} \cdot \mathbf{u})\mathbf{u}$ and $\mathbf{a}_{\perp} = \mathbf{a} - \mathbf{a}_{\parallel}$. Then

$$\det(A) = -\gamma^4(\mathbf{a}_{\perp}^2 + \gamma^2\mathbf{a}_{\parallel}^2). \quad (5.18)$$

The product U^*A has zero scalar part, real vector $\gamma^3\mathbf{a}$, and imaginary vector $-\gamma^3\mathbf{v} \times \mathbf{a}$; while UA has real scalar $2\gamma^5\mathbf{v} \cdot \mathbf{a}$, real vector $\gamma^3\mathbf{a} + 2\gamma^5(\mathbf{v} \cdot \mathbf{a})\mathbf{v}$, and imaginary vector $\gamma^3\mathbf{v} \times \mathbf{a}$.

5.2 Free-particle action

The free-particle action and Lagrangian are

$$\int L dt = -m \int ds, \quad L = -m\sqrt{1 - \mathbf{v}^2} = -\frac{m}{\gamma}. \quad (5.19)$$

For a general Lagrangian $L(t, \mathbf{r}, \mathbf{v})$,

$$\frac{d}{dt} \partial_{\mathbf{v}} L = \partial_{\mathbf{r}} L, \quad H = \mathbf{v} \cdot \partial_{\mathbf{v}} L - L, \quad \frac{dH}{dt} = -\partial_t L. \quad (5.20)$$

Thus spatial translation invariance conserves canonical momentum and time translation invariance conserves the Hamiltonian. For the free-particle Lagrangian,

$$\partial_{\mathbf{v}} L = m\gamma \mathbf{v}, \quad (5.21)$$

$$H = m\gamma \mathbf{v}^2 + \frac{m}{\gamma} = m\gamma(\mathbf{v}^2 + \gamma^{-2}) = m\gamma. \quad (5.22)$$

At low velocity, the binomial expansions are

$$\sqrt{1 - \mathbf{v}^2} = 1 - \frac{1}{2}\mathbf{v}^2 + O(v^4), \quad \gamma = 1 + \frac{1}{2}\mathbf{v}^2 + O(v^4). \quad (5.23)$$

Multiplying the first expansion by $-m$ gives

$$L = -m\sqrt{1 - \mathbf{v}^2} = -m + \frac{1}{2}m\mathbf{v}^2 + O(v^4), \quad (5.24)$$

while the second gives

$$H = m\gamma = m + \frac{1}{2}m\mathbf{v}^2 + O(v^4). \quad (5.25)$$

Thus, apart from the constant rest-energy term, both recover the nonrelativistic kinetic energy $\frac{1}{2}m\mathbf{v}^2$.

6 Electromagnetism

6.1 Field equation and potentials

Introduce the paravector differential operator, electromagnetic field, and source

$$\partial = \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \quad F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad D = \rho + \mathbf{J} \cdot \boldsymbol{\sigma}. \quad (6.1)$$

All four Maxwell equations are the single equation

$$\boxed{\partial F = D^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}.} \quad (6.2)$$

Expanding the left side by component gives

$$\partial F = \underbrace{\partial_{\mathbf{r}} \cdot \mathbf{E}}_{\text{real scalar}} + \underbrace{i\partial_{\mathbf{r}} \cdot \mathbf{B}}_{\text{imaginary scalar}} \quad (6.3)$$

$$+ \underbrace{(\partial_t \mathbf{E} - \partial_{\mathbf{r}} \times \mathbf{B}) \cdot \boldsymbol{\sigma}}_{\text{real vector}} \quad (6.4)$$

$$+ \underbrace{i(\partial_t \mathbf{B} + \partial_{\mathbf{r}} \times \mathbf{E}) \cdot \boldsymbol{\sigma}}_{\text{imaginary vector}}. \quad (6.5)$$

Equating these four components with those of $\rho - \mathbf{J} \cdot \boldsymbol{\sigma}$ gives, respectively,

$$\text{real scalar: } \partial_{\mathbf{r}} \cdot \mathbf{E} = \rho, \quad (6.6)$$

$$\text{imaginary scalar: } \partial_{\mathbf{r}} \cdot \mathbf{B} = 0, \quad (6.7)$$

$$\text{real vector: } \partial_{\mathbf{r}} \times \mathbf{B} = \mathbf{J} + \partial_t \mathbf{E}, \quad (6.8)$$

$$\text{imaginary vector: } \partial_{\mathbf{r}} \times \mathbf{E} = -\partial_t \mathbf{B}. \quad (6.9)$$

Because

$$\partial \partial^* = \partial^* \partial = \partial_t^2 - \partial_{\mathbf{r}}^2 =: \square, \quad (6.10)$$

applying ∂^* to (6.2) gives

$$\square F = \partial^* D^* \quad (6.11)$$

$$= (\partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J}) + (-\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho + i \partial_{\mathbf{r}} \times \mathbf{J}) \cdot \boldsymbol{\sigma}. \quad (6.12)$$

The real-scalar component is charge conservation, and the real- and imaginary-vector components are the electric and magnetic wave equations:

$$\text{real scalar: } \partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J} = 0, \quad (6.13)$$

$$\text{real vector: } \square \mathbf{E} = -\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho, \quad (6.14)$$

$$\text{imaginary vector: } \square \mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{J}. \quad (6.15)$$

For the potential $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$,

$$\partial \Phi = \underbrace{\partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}}_{\text{real scalar}} + \underbrace{(\partial_t \mathbf{A} + \partial_{\mathbf{r}} V) \cdot \boldsymbol{\sigma}}_{\text{real vector}} \quad (6.16)$$

$$+ i \underbrace{(\partial_{\mathbf{r}} \times \mathbf{A}) \cdot \boldsymbol{\sigma}}_{\text{imaginary vector}}. \quad (6.17)$$

Define the scalar component

$$S := \text{sc}(\partial \Phi) = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}. \quad (6.18)$$

Then $\partial \Phi = S + F^*$: its real-vector and imaginary-vector components give

$$\mathbf{E} = -\partial_t \mathbf{A} - \partial_{\mathbf{r}} V, \quad \mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{A}. \quad (6.19)$$

Also, $F = \partial^* \Phi^* - S$, so

$$\partial F = \partial \partial^* \Phi^* - \partial S = \square \Phi^* - \partial S \quad (6.20)$$

$$= (\square V - \partial_t S) - (\square \mathbf{A} + \partial_{\mathbf{r}} S) \cdot \boldsymbol{\sigma}. \quad (6.21)$$

Comparing its real-scalar and real-vector components with $\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$ gives

$$\rho = \square V - \partial_t S, \quad \mathbf{J} = \square \mathbf{A} + \partial_{\mathbf{r}} S. \quad (6.22)$$

For a scalar field λ , the gauge transformation is

$$\Phi' = \Phi + \partial^* \lambda, \quad (6.23)$$

$$\partial \Phi' = \partial \Phi + \partial \partial^* \lambda = \partial \Phi + \square \lambda. \quad (6.24)$$

Because $\square \lambda$ is scalar, only the scalar component changes:

$$S' = S + \square \lambda, \quad \text{vec}(\partial \Phi') = \text{vec}(\partial \Phi). \quad (6.25)$$

Thus F is unchanged. In the Lorenz gauge $S = 0$, equation (6.22) reduces to

$$\rho = \square V, \quad \mathbf{J} = \square \mathbf{A}. \quad (6.26)$$

6.2 Invariants, energy, and force

For $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$, two simple products contain the standard electromagnetic invariants and energy flow:

$$\frac{1}{2}F^2 = \frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2) + i\mathbf{E} \cdot \mathbf{B}, \quad (6.27)$$

$$\frac{1}{2}FF^H = \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (6.28)$$

Thus the scalar and vector parts of the second expression are the field energy density and Poynting vector. The covariant Lorentz force is

$$m \frac{dU}{ds} = q \operatorname{re}(FU) = \frac{q}{2}(FU + UF^H), \quad (6.29)$$

which reproduces $d(m\gamma)/dt = q\mathbf{v} \cdot \mathbf{E}$ and $d\mathbf{P}/dt = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$.

6.3 Derivation of force from the potential and action

The interacting particle action is

$$\int L dt = -m \int ds - q \int \operatorname{sc}(\Phi^* U) ds \quad (6.30)$$

$$= \int \left(-\frac{m}{\gamma} - q \operatorname{sc}(\Phi^* U/\gamma) \right) dt \quad (6.31)$$

$$= \int \left(-\frac{m}{\gamma} - q \operatorname{sc}((V - \mathbf{A} \cdot \boldsymbol{\sigma})(1 + \mathbf{v} \cdot \boldsymbol{\sigma})) \right) dt \quad (6.32)$$

$$= \int \left(-\frac{m}{\gamma} - qV + q\mathbf{v} \cdot \mathbf{A} \right) dt. \quad (6.33)$$

Thus

$$L = -\frac{m}{\gamma} - q(V - \mathbf{v} \cdot \mathbf{A}), \quad \partial_{\mathbf{v}} L = m\gamma\mathbf{v} + q\mathbf{A} = \mathbf{P} + q\mathbf{A}. \quad (6.34)$$

The Euler–Lagrange terms are

$$\frac{d}{dt} \partial_{\mathbf{v}} L = \frac{d\mathbf{P}}{dt} + q\partial_t \mathbf{A} + q(\mathbf{v} \cdot \partial_{\mathbf{r}}) \mathbf{A}, \quad (6.35)$$

$$\partial_{\mathbf{r}} L = -q\partial_{\mathbf{r}} V + q\partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}). \quad (6.36)$$

Using $(\mathbf{v} \cdot \partial_{\mathbf{r}}) \mathbf{A} = \partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}) - \mathbf{v} \times (\partial_{\mathbf{r}} \times \mathbf{A})$ gives

$$\frac{d\mathbf{P}}{dt} = -q\partial_{\mathbf{r}} V - q\partial_t \mathbf{A} + q(\partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}) - (\mathbf{v} \cdot \partial_{\mathbf{r}}) \mathbf{A}) \quad (6.37)$$

$$= q(-\partial_{\mathbf{r}} V - \partial_t \mathbf{A} + \mathbf{v} \times (\partial_{\mathbf{r}} \times \mathbf{A})) \quad (6.38)$$

$$= q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \quad (6.39)$$

The Hamiltonian follows with the gauge-dependent terms cancelling:

$$H = \mathbf{v} \cdot \partial_{\mathbf{v}} L - L \quad (6.40)$$

$$= m\gamma\mathbf{v}^2 + q\mathbf{v} \cdot \mathbf{A} + \frac{m}{\gamma} + qV - q\mathbf{v} \cdot \mathbf{A} \quad (6.41)$$

$$= m\gamma(\mathbf{v}^2 + \gamma^{-2}) + qV = m\gamma + qV. \quad (6.42)$$

Finally, $dH/dt = -\partial_t L$ gives

$$\frac{d(m\gamma)}{dt} + q\partial_t V + q\mathbf{v} \cdot \partial_{\mathbf{r}} V = q\partial_t V - q\mathbf{v} \cdot \partial_t \mathbf{A}, \quad (6.43)$$

$$\frac{d(m\gamma)}{dt} = q\mathbf{v} \cdot (-\partial_{\mathbf{r}} V - \partial_t \mathbf{A}) = q\mathbf{v} \cdot \mathbf{E}. \quad (6.44)$$

To see equation (6.29) componentwise, expand

$$\frac{FU}{\gamma} = \mathbf{v} \cdot \mathbf{E} + i\mathbf{v} \cdot \mathbf{B} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma} + i(\mathbf{B} + \mathbf{E} \times \mathbf{v}) \cdot \boldsymbol{\sigma}. \quad (6.45)$$

Taking the real paravector part removes the two imaginary components.

6.4 Energy balance

Let

$$T := \frac{1}{2} F F^H = T_s + \mathbf{T}_v \cdot \boldsymbol{\sigma}, \quad T_s = \frac{1}{2} (\mathbf{E}^2 + \mathbf{B}^2), \quad \mathbf{T}_v = \mathbf{E} \times \mathbf{B}. \quad (6.46)$$

Then the scalar part of ∂T is $\partial_t T_s + \partial_{\mathbf{r}} \cdot \mathbf{T}_v$. The source products needed for local balance have components

$$F D^* : -\mathbf{E} \cdot \mathbf{J} - i\mathbf{B} \cdot \mathbf{J} + (\rho \mathbf{E} + \mathbf{B} \times \mathbf{J}) \cdot \boldsymbol{\sigma} + i(\rho \mathbf{B} - \mathbf{E} \times \mathbf{J}) \cdot \boldsymbol{\sigma}, \quad (6.47)$$

$$D^* F : -\mathbf{E} \cdot \mathbf{J} - i\mathbf{B} \cdot \mathbf{J} + (\rho \mathbf{E} - \mathbf{B} \times \mathbf{J}) \cdot \boldsymbol{\sigma} + i(\rho \mathbf{B} + \mathbf{E} \times \mathbf{J}) \cdot \boldsymbol{\sigma}. \quad (6.48)$$

Since the derivative is not an ordinary scalar derivation,

$$\partial(F^2) = D^* F + F D^* + [\boldsymbol{\sigma}, F] \cdot \partial_{\mathbf{r}} F. \quad (6.49)$$

Writing $F = \mathbf{F} \cdot \boldsymbol{\sigma}$, its last term is

$$-2i\mathbf{F} \cdot (\partial_{\mathbf{r}} \times \mathbf{F}) - 2((\partial_{\mathbf{r}} \cdot \mathbf{F})\mathbf{F} - \frac{1}{2}\partial_{\mathbf{r}}(F^2)) \cdot \boldsymbol{\sigma}. \quad (6.50)$$

7 Relativistic wave equations

7.1 Paravector inner products and expectations

A paravector amplitude ψ maps to the complex matrix $\mathbf{M}_{\psi} = \text{mat}(\psi)$. Therefore $\text{mat}(\psi^H \psi) = \mathbf{M}_{\psi}^H \mathbf{M}_{\psi}$, which is generally a Hermitian matrix rather than a scalar.

For paravector amplitudes ψ, ψ' and a paravector Z , use the Hilbert–Schmidt pairing to define

$$\langle \psi' | \psi \rangle := \text{sc}(\psi'^H \psi) = \frac{1}{2} \text{tr}(\mathbf{M}_{\psi'}^H \mathbf{M}_{\psi}), \quad (7.1)$$

$$\langle \psi' | Z | \psi \rangle := \text{sc}(\psi'^H Z \psi) = \frac{1}{2} \text{tr}(\mathbf{M}_{\psi'}^H \text{mat}(Z) \mathbf{M}_{\psi}). \quad (7.2)$$

The first is the $Z = 1$ case of the second, and $\langle \psi | \psi \rangle = \|\psi\|^2$. Thus normalization is $\text{sc}(\psi^H \psi) = 1$, not $\psi^H \psi = 1$. For an observable $Z^H = Z$ acting from the left,

$$\langle Z \rangle_{\psi} := \frac{\langle \psi | Z | \psi \rangle}{\langle \psi | \psi \rangle} = \frac{\text{sc}(\psi^H Z \psi)}{\text{sc}(\psi^H \psi)}. \quad (7.3)$$

For normalized ψ , this reduces to $\langle Z \rangle_\psi = \langle \psi | Z | \psi \rangle$.

For a paravector T with $\psi' = T\psi$,

$$\langle Z \rangle_{\psi'} = \frac{\text{sc}(\psi^H T^H Z T \psi)}{\text{sc}(\psi^H T^H T \psi)}. \quad (7.4)$$

For a unitary rotation the denominator is unchanged. A finite boost is not unitary, so this positive equal-time norm is not itself Lorentz invariant; the covariant Dirac normalization instead comes from the conserved Dirac current on a spacelike hypersurface.

7.2 Physical spinors and minimal ideals

A physical two-component Pauli spinor occupies a minimal left ideal. For any unit direction $\mathbf{u}$, define the algebra and matrix projectors

$$\Pi_\pm(\mathbf{u}) := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \quad \mathbf{\Pi}_\pm(\mathbf{u}) := \frac{1}{2}(\mathbf{I}_2 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (7.5)$$

They satisfy

$$\mathbf{\Pi}_\pm(\mathbf{u})^2 = \mathbf{\Pi}_\pm(\mathbf{u}), \quad \mathbf{\Pi}_+(\mathbf{u})\mathbf{\Pi}_-(\mathbf{u}) = 0, \quad \mathbf{\Pi}_+(\mathbf{u}) + \mathbf{\Pi}_-(\mathbf{u}) = \mathbf{I}_2, \quad (7.6)$$

$$\Pi_\pm(\mathbf{u})^H = \Pi_\pm(\mathbf{u}), \quad \Pi_\pm(\mathbf{u})^* = \Pi_\mp(\mathbf{u}), \quad (\mathbf{u} \cdot \boldsymbol{\sigma})\Pi_\pm(\mathbf{u}) = \pm \mathbf{\Pi}_\pm(\mathbf{u}), \quad (7.7)$$

and hence $\mathbf{u} \cdot \boldsymbol{\sigma} = \mathbf{\Pi}_+(\mathbf{u}) - \mathbf{\Pi}_-(\mathbf{u})$.

Choose a fixed reference direction $\mathbf{u}_0$ and project on the right:

$$\psi' := \psi \Pi_+(\mathbf{u}_0), \quad \mathbf{M}_{\psi'} := \text{mat}(\psi') = \mathbf{M}_\psi \mathbf{\Pi}_+(\mathbf{u}_0). \quad (7.8)$$

Let $\mathbf{e}_+(\mathbf{u}_0)$ be a normalized positive eigenvector,

$$(\mathbf{u}_0 \cdot \boldsymbol{\sigma})\mathbf{e}_+(\mathbf{u}_0) = \mathbf{e}_+(\mathbf{u}_0), \quad \mathbf{e}_+(\mathbf{u}_0)^H \mathbf{e}_+(\mathbf{u}_0) = 1. \quad (7.9)$$

A convenient phase choice, valid for $u_{0,3} \neq -1$, is

$$\mathbf{e}_+(\mathbf{u}_0) = \frac{\{1 + u_{0,3}, u_{0,1} + i u_{0,2}\}^T}{\sqrt{2(1 + u_{0,3})}}; \quad \mathbf{e}_+(-\mathbf{e}_3) = \{0, 1\}^T. \quad (7.10)$$

Define the conventional column spinor $\chi \in \mathbb{C}^2$ by

$$\mathbf{\Pi}_+(\mathbf{u}_0) = \mathbf{e}_+(\mathbf{u}_0)\mathbf{e}_+(\mathbf{u}_0)^H, \quad \chi := \mathbf{M}_\psi \mathbf{e}_+(\mathbf{u}_0). \quad (7.11)$$

The algebra, rank-one matrix, and column forms of the state are related by

$$\chi = \mathbf{M}_{\psi'} \mathbf{e}_+(\mathbf{u}_0), \quad \mathbf{M}_{\psi'} = \chi \mathbf{e}_+(\mathbf{u}_0)^H. \quad (7.12)$$

Thus ψ' and χ contain the same two complex components. $\mathbf{M}_{\psi'}$ has rank one and has χ as its first column precisely when the basis begins with $\mathbf{e}_+(\mathbf{u}_0)$. Conversely, $\chi \mathbf{e}_+(\mathbf{u}_0)^H$ uniquely determines ψ' . The column χ is a 2×1 vector, so $\chi^H \chi$ is scalar, and

$$\chi^H \chi = \text{tr}(\mathbf{M}_{\psi'}^H \mathbf{M}_{\psi'}) = 2 \text{sc}(\psi'^H \psi'), \quad (7.13)$$

$$\chi^H \text{mat}(Z) \chi = \text{tr}(\mathbf{M}_{\psi'}^H \text{mat}(Z) \mathbf{M}_{\psi'}) = 2 \text{sc}(\psi'^H Z \psi'). \quad (7.14)$$

The factor two cancels in normalized expectations. An unrestricted paravector matrix contains two Pauli-spinor columns; right projection by Π_+ , or by $\mathbf{\Pi}_+$ in matrix form, selects one. Applied to both components of the Dirac block below, it selects one conventional four-component Dirac column.

7.3 Klein–Gordon equation

With $E = i\partial_t$ and $\mathbf{P} = -i\partial_{\mathbf{r}}$, the mass-shell relation becomes the scalar wave equation

$$(\square + m^2)\Psi = 0. \quad (7.15)$$

Writing $\Omega = \sqrt{m^2 - \partial_{\mathbf{r}}^2}$ gives the frequency decomposition

$$(\partial_t \pm i\Omega)\Psi_{\pm} = 0, \quad \Psi = \Psi_+ + \Psi_-. \quad (7.16)$$

The square-root operator is understood spectrally: on a plane wave with wave vector $\mathbf{k}$, Ω multiplies by $\sqrt{m^2 + \mathbf{k}^2}$.

Minimal electromagnetic coupling replaces the operators by

$$E = i\partial_t - qV, \quad \mathbf{P} = -i\partial_{\mathbf{r}} - q\mathbf{A}, \quad (7.17)$$

so that $(-E^2 + \mathbf{P}^2 + m^2)\Psi = 0$.

7.4 Dirac equation in block form

For a paravector Z , define

$$\mathbf{W}(Z) := \begin{pmatrix} 0 & Z \\ \text{adj}(Z) & 0 \end{pmatrix}. \quad (7.18)$$

It satisfies

$$\mathbf{W}(Z)^2 = \text{diag}(Z \text{adj}(Z), \text{adj}(Z) Z), \quad \mathbf{W}(Z)^* = \mathbf{W}(Z^*), \quad \mathbf{W}(Z)^{\mathbf{H}} = \mathbf{W}(Z^{\mathbf{H}}). \quad (7.19)$$

Since $\mathbf{W}(i\partial)^2 = -\square \mathbf{I}$, the first-order equation

$$\boxed{(\mathbf{W}(i\partial) - m\mathbf{I})\Psi = 0} \quad (7.20)$$

squares to the Klein–Gordon equation. For $\Psi = \{\Psi_1, \Psi_2\}^{\mathbf{T}}$ it reads

$$i\partial\Psi_2 - m\Psi_1 = 0, \quad i\partial^*\Psi_1 - m\Psi_2 = 0. \quad (7.21)$$

The characteristic equation is $(s + m)^2 + \square = 0$, hence $s = -m \pm i\sqrt{\square}$. With a potential, the gauge-covariant form is

$$(\mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I})\Psi = 0. \quad (7.22)$$

It is invariant under $\Phi' = \Phi + \partial^*\lambda$ and $\Psi = e^{iq\lambda}\Psi'$.

7.5 Schrödinger–Pauli limit

Removing the rest-energy oscillation and expanding at low momentum gives the Pauli equation

$$i\partial_t\Psi = \left(\frac{(-i\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV - \frac{q}{2m}\mathbf{B} \cdot \boldsymbol{\sigma} \right) \Psi. \quad (7.23)$$

Using these projectors, $\Psi = \Psi_+ + \Psi_-$ with $\Psi_{\pm} = \Pi_{\pm}(\mathbf{u})\Psi$. For a uniform static magnetic field, $\mathbf{u} = \mathbf{B}/\|\mathbf{B}\|$, the sectors decouple:

$$i\partial_t\Psi_{\pm} = \left(\frac{\mathbf{P}^2 \mp q\|\mathbf{B}\|}{2m} + qV \right) \Psi_{\pm}. \quad (7.24)$$

For a varying field direction the sectors are coupled. With H denoting the Pauli Hamiltonian, direct differentiation gives

$$\mathbf{i}\partial_t \Psi_{\pm} = \mathbf{i}(\partial_t \Pi_{\pm}) \Psi + \Pi_{\pm} H \Psi \quad (7.25)$$

$$= H \Psi_{\pm} + (\mathbf{i}\partial_t \Pi_{\pm} + [\Pi_{\pm}, H]) \Psi. \quad (7.26)$$

Because the field term commutes with its own instantaneous projectors, this is

$$\mathbf{i}\partial_t \Psi_{\pm} = H \Psi_{\pm} + S_{\pm}, \quad S_{\pm} = \left(\mathbf{i}\partial_t \Pi_{\pm} + \frac{[\mathbf{P}^2, \Pi_{\pm}]}{2m} \right) \Psi. \quad (7.27)$$

7.6 Interacting operators and factorization

The covariant energy and momentum operators do not commute. Direct application to a test wavefunction gives

$$[E, P_m] = \mathbf{i}qE_m, \quad [P_m, P_n] = \mathbf{i}q(\partial_m A_n - \partial_n A_m), \quad \mathbf{P} \times \mathbf{P} = \mathbf{i}q\mathbf{B}. \quad (7.28)$$

Therefore

$$(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - \mathbf{i}qF, \quad (7.29)$$

$$(E - \boldsymbol{\sigma} \cdot \mathbf{P})(E + \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - \mathbf{i}qF^*, \quad (7.30)$$

$$E + \boldsymbol{\sigma} \cdot \mathbf{P} = \mathbf{i}\partial^* - q\Phi, \quad (7.31)$$

$$E - \boldsymbol{\sigma} \cdot \mathbf{P} = \mathbf{i}\partial - q\Phi^*. \quad (7.32)$$

Expanding the minimally coupled Klein–Gordon operator gives the fully explicit result

$$(\square + \mathbf{i}qS - q^2 \det(\Phi) + m^2 + 2\mathbf{i}q(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}})) \Psi = 0. \quad (7.33)$$

The two orderings satisfy

$$0 = ((\partial + \mathbf{i}q\Phi^*)(\partial^* + \mathbf{i}q\Phi) + m^2) \Psi_1 = (-E^2 + \mathbf{P}^2 + m^2 + \mathbf{i}qF^*) \Psi_1, \quad (7.34)$$

$$0 = ((\partial^* + \mathbf{i}q\Phi)(\partial + \mathbf{i}q\Phi^*) + m^2) \Psi_2 = (-E^2 + \mathbf{P}^2 + m^2 + \mathbf{i}qF) \Psi_2. \quad (7.35)$$

These follow by multiplying the Dirac operator by its mass conjugate. In detail, if $\mathbf{D} = \mathbf{W}(\mathbf{i}\partial - q\Phi^*) - m\mathbf{I}$ and $\mathbf{D}' = \mathbf{W}(\mathbf{i}\partial - q\Phi^*) + m\mathbf{I}$, then

$$\mathbf{D}'\mathbf{D} = -\text{diag}((\partial + \mathbf{i}q\Phi^*)(\partial^* + \mathbf{i}q\Phi) + m^2, (\partial^* + \mathbf{i}q\Phi)(\partial + \mathbf{i}q\Phi^*) + m^2). \quad (7.36)$$

For a paravector wavefunction $\psi = \psi_s + \boldsymbol{\psi}_v \cdot \boldsymbol{\sigma}$,

$$\partial\psi = (\partial_t\psi_s + \partial_{\mathbf{r}} \cdot \boldsymbol{\psi}_v) + (\partial_t\boldsymbol{\psi}_v + \partial_{\mathbf{r}}\psi_s + \mathbf{i}\partial_{\mathbf{r}} \times \boldsymbol{\psi}_v) \cdot \boldsymbol{\sigma}, \quad (7.37)$$

$$\partial^*\psi = (\partial_t\psi_s - \partial_{\mathbf{r}} \cdot \boldsymbol{\psi}_v) + (\partial_t\boldsymbol{\psi}_v - \partial_{\mathbf{r}}\psi_s - \mathbf{i}\partial_{\mathbf{r}} \times \boldsymbol{\psi}_v) \cdot \boldsymbol{\sigma}, \quad (7.38)$$

$$\Phi\psi = (V\psi_s + \mathbf{A} \cdot \boldsymbol{\psi}_v) + (V\boldsymbol{\psi}_v + \psi_s\mathbf{A} + \mathbf{i}\mathbf{A} \times \boldsymbol{\psi}_v) \cdot \boldsymbol{\sigma}, \quad (7.39)$$

$$\Phi^*\psi = (V\psi_s - \mathbf{A} \cdot \boldsymbol{\psi}_v) + (V\boldsymbol{\psi}_v - \psi_s\mathbf{A} - \mathbf{i}\mathbf{A} \times \boldsymbol{\psi}_v) \cdot \boldsymbol{\sigma}. \quad (7.40)$$

Writing $F = \mathbf{F} \cdot \boldsymbol{\sigma}$,

$$\text{sc}(\mathbf{i}qF\psi) = \mathbf{i}q\mathbf{F} \cdot \boldsymbol{\psi}_v, \quad \text{vec}(\mathbf{i}qF\psi) = \mathbf{i}q\psi_s\mathbf{F} - q\mathbf{F} \times \boldsymbol{\psi}_v, \quad (7.41)$$

$$\text{sc}(\mathbf{i}qF^*\psi) = -\mathbf{i}q\mathbf{F}^* \cdot \boldsymbol{\psi}_v, \quad \text{vec}(\mathbf{i}qF^*\psi) = -\mathbf{i}q\psi_s\mathbf{F}^* + q\mathbf{F}^* \times \boldsymbol{\psi}_v. \quad (7.42)$$

7.7 Low-energy derivation

For spin zero, write $\Psi' = e^{-imt}\Psi$. The exact equation becomes

$$(-E^2 - 2mE + \mathbf{P}^2)\Psi = 0, \quad E = \frac{\mathbf{P}^2 - E^2}{2m} = \frac{\mathbf{P}^2}{2m} - \frac{\mathbf{P}^4}{8m^3} + \dots \quad (7.43)$$

The leading term is the Schrödinger equation

$$i\partial_t\Psi = \left(\frac{(-i\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV \right) \Psi. \quad (7.44)$$

For spin one-half, the two rest-mass-stripped Dirac equations are

$$(E + m - \mathbf{P} \cdot \boldsymbol{\sigma})\Psi_2 - m\Psi_1 = 0, \quad (7.45)$$

$$(E + m + \mathbf{P} \cdot \boldsymbol{\sigma})\Psi_1 - m\Psi_2 = 0. \quad (7.46)$$

Eliminating Ψ_1 gives

$$(E^2 + 2mE - \mathbf{P}^2 + q\mathbf{B} \cdot \boldsymbol{\sigma} - iq\mathbf{E} \cdot \boldsymbol{\sigma})\Psi_2 = 0, \quad (7.47)$$

and iteration yields $E = (\mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma})/(2m) + O(m^{-2})$, which is equation (7.23).

7.8 Spin, ladder operators, and energy sectors

The left measurement projection along $\mathbf{u}$ is distinct from the fixed right ideal projection along $\mathbf{u}_0$ defined above. For the ideal matrix $\mathbf{M}_{\psi'}$ and its nonzero column χ , define

$$\mathbf{M}_{\psi', \pm} := \boldsymbol{\Pi}_{\pm}(\mathbf{u})\mathbf{M}_{\psi'}, \quad \chi_{\pm} := \boldsymbol{\Pi}_{\pm}(\mathbf{u})\chi. \quad (7.48)$$

The components are orthogonal and complete:

$$\chi = \chi_+ + \chi_-, \quad \chi_+^H \chi_- = 0, \quad \chi_-^H \chi_+ = 0. \quad (7.49)$$

For a normalized state, define its real Bloch vector by

$$s_n := \frac{\chi^H \boldsymbol{\sigma}_n \chi}{\chi^H \chi}, \quad \mathbf{s} := \{s_1, s_2, s_3\}, \quad \|\mathbf{s}\|^2 = 1. \quad (7.50)$$

The probabilities for spin along and against $\mathbf{u}$ are

$$p_{\pm}(\mathbf{u}) := \frac{\chi_{\pm}^H \chi_{\pm}}{\chi^H \chi} = \frac{\chi^H \boldsymbol{\Pi}_{\pm}(\mathbf{u}) \chi}{\chi^H \chi} = \frac{1}{2}(1 \pm \mathbf{u} \cdot \mathbf{s}). \quad (7.51)$$

For two unit directions $\mathbf{u}, \mathbf{u}'$, the product and commutator are

$$(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) = \mathbf{u} \cdot \mathbf{u}' + i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (7.52)$$

$$[\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}] = 2i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}. \quad (7.53)$$

Moreover,

$$(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) = (2(\mathbf{u} \cdot \mathbf{u}')\mathbf{u} - \mathbf{u}') \cdot \boldsymbol{\sigma}. \quad (7.54)$$

Define $\mathbf{u}'_{\parallel} = (\mathbf{u}' \cdot \mathbf{u})\mathbf{u}$ and $\mathbf{u}'_{\perp} = \mathbf{u}' - \mathbf{u}'_{\parallel}$. The same-side and cross-side projector identities are

$$\mathbf{\Pi}_{\pm}(\mathbf{u})(\mathbf{u}' \cdot \boldsymbol{\sigma})\mathbf{\Pi}_{\pm}(\mathbf{u}) = \pm(\mathbf{u} \cdot \mathbf{u}')\mathbf{\Pi}_{\pm}(\mathbf{u}), \quad (7.55)$$

$$\mathbf{\Pi}_{\pm}(\mathbf{u})(\mathbf{u}' \cdot \boldsymbol{\sigma})\mathbf{\Pi}_{\mp}(\mathbf{u}) = \frac{1}{2}(\mathbf{u}'_{\perp} \pm i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma} \mathbf{\Pi}_{\mp}(\mathbf{u}). \quad (7.56)$$

For a state aligned with $\mathbf{u}$ and $\mathbf{u} \cdot \mathbf{u}' = \cos \theta$, the measurement probabilities along $\mathbf{u}'$ are

$$p_{+}(\mathbf{u}') = \frac{1}{2}(1 + \mathbf{u} \cdot \mathbf{u}') = \cos^2(\theta/2), \quad p_{-}(\mathbf{u}') = \frac{1}{2}(1 - \mathbf{u} \cdot \mathbf{u}') = \sin^2(\theta/2). \quad (7.57)$$

If $\mathbf{u}'$ is obtained by rotating $\mathbf{u}$ through ϕ about $\mathbf{n}$, then

$$\mathbf{u} \cdot \mathbf{u}' = (\mathbf{n} \cdot \mathbf{u})^2 + (1 - (\mathbf{n} \cdot \mathbf{u})^2) \cos \phi. \quad (7.58)$$

Thus $\theta = \phi$ when $\mathbf{n} \cdot \mathbf{u} = 0$.

For fluctuations $d\mathbf{u}$ and $d\mathbf{u}'$, the squared determinant identity is

$$\det((d\mathbf{u} \cdot \boldsymbol{\sigma})(d\mathbf{u}' \cdot \boldsymbol{\sigma})) = (d\mathbf{u} \cdot d\mathbf{u}')^2 + \frac{1}{4} \|[d\mathbf{u} \cdot \boldsymbol{\sigma}, d\mathbf{u}' \cdot \boldsymbol{\sigma}]\|^2. \quad (7.59)$$

For orthogonal fluctuations this reduces to

$$\det((d\mathbf{u} \cdot \boldsymbol{\sigma})(d\mathbf{u}' \cdot \boldsymbol{\sigma})) = \frac{1}{4} \|[d\mathbf{u} \cdot \boldsymbol{\sigma}, d\mathbf{u}' \cdot \boldsymbol{\sigma}]\|^2. \quad (7.60)$$

About the third axis, define

$$\mathbf{a}_{\pm} := \frac{1}{2}(\boldsymbol{\sigma}_1 \pm i\boldsymbol{\sigma}_2), \quad \mathbf{\Pi}_{\pm} := \frac{1}{2}(\mathbf{I}_2 \pm \boldsymbol{\sigma}_3). \quad (7.61)$$

Then

$$[\boldsymbol{\sigma}_3, \mathbf{a}_{\pm}] = \pm 2\mathbf{a}_{\pm}, \quad \mathbf{a}_{\pm}^2 = 0, \quad \mathbf{a}_{\pm}\mathbf{a}_{\mp} = \mathbf{\Pi}_{\pm}, \quad (7.62)$$

$$\boldsymbol{\sigma}_1 = \mathbf{a}_{+} + \mathbf{a}_{-}, \quad \boldsymbol{\sigma}_2 = (\mathbf{a}_{+} - \mathbf{a}_{-})/i, \quad \boldsymbol{\sigma}_3 = \mathbf{\Pi}_{+} - \mathbf{\Pi}_{-}. \quad (7.63)$$

Their projector action is

$$\mathbf{a}_{+}\mathbf{\Pi}_{+} = 0, \quad \mathbf{a}_{+}\mathbf{\Pi}_{-} = \mathbf{a}_{+}, \quad \mathbf{\Pi}_{+}\mathbf{a}_{+} = \mathbf{a}_{+}, \quad \mathbf{\Pi}_{-}\mathbf{a}_{+} = 0, \quad (7.64)$$

$$\mathbf{a}_{-}\mathbf{\Pi}_{-} = 0, \quad \mathbf{a}_{-}\mathbf{\Pi}_{+} = \mathbf{a}_{-}, \quad \mathbf{\Pi}_{-}\mathbf{a}_{-} = \mathbf{a}_{-}, \quad \mathbf{\Pi}_{+}\mathbf{a}_{-} = 0. \quad (7.65)$$

Finally, on the mass shell let

$$U = \frac{E + \mathbf{P} \cdot \boldsymbol{\sigma}}{m}, \quad \boldsymbol{\Lambda}_{\pm} = \frac{1}{2}(\mathbf{I} \pm \mathbf{W}(U)). \quad (7.66)$$

Since $\det(U) = 1$ and $\mathbf{W}(U)^2 = \mathbf{I}$, the positive- and negative-energy projectors satisfy

$$\boldsymbol{\Lambda}_{\pm}^2 = \boldsymbol{\Lambda}_{\pm}, \quad \boldsymbol{\Lambda}_{+}\boldsymbol{\Lambda}_{-} = 0, \quad \boldsymbol{\Lambda}_{+} + \boldsymbol{\Lambda}_{-} = \mathbf{I}. \quad (7.67)$$

Lift the spin projectors to the Dirac block space by

$$\mathbf{\Pi}_{j,4} := \text{diag}(\mathbf{\Pi}_j, \mathbf{\Pi}_j), \quad j = \pm. \quad (7.68)$$

When $[\boldsymbol{\Lambda}_i, \mathbf{\Pi}_{j,4}] = 0$, as for a compatible spin quantization axis, their products are joint energy-spin projectors and resolve the physical Dirac column selected in Section 7.2 into four sectors:

$$\boldsymbol{\Psi} = \sum_{i=\pm} \sum_{j=\pm} \boldsymbol{\Psi}_{ij}, \quad \boldsymbol{\Psi}_{ij} = \boldsymbol{\Lambda}_i \mathbf{\Pi}_{j,4} \boldsymbol{\Psi}. \quad (7.69)$$

Without this commutation condition the displayed sum remains a sequential decomposition, but the factors are not joint orthogonal projectors.

8 Rotations and Lorentz boosts

8.1 Rotors

A rotation through angle ϕ about unit axis $\mathbf{u}$ is generated by

$$R = \exp\left(\frac{\phi}{2} \mathbf{u} \cdot \mathbf{i}\boldsymbol{\sigma}\right) = \cos \frac{\phi}{2} + \mathbf{u} \cdot \mathbf{i}\boldsymbol{\sigma} \sin \frac{\phi}{2}, \quad R^{-1} = R^H. \quad (8.1)$$

The sandwich action $X' = R^H X R$ fixes the scalar component and rotates the spatial vector according to Rodrigues' formula,

$$\mathbf{r}' = \mathbf{r} \cos \phi + (\mathbf{u} \times \mathbf{r}) \sin \phi + (\mathbf{u} \cdot \mathbf{r})(1 - \cos \phi)\mathbf{u}. \quad (8.2)$$

The half-angle explains the spinorial sign change $R(\phi + 2\pi) = -R(\phi)$. In the physical column representation, let $\mathbf{R} := \text{mat}(R)$ and define the pure-state density matrix

$$\rho_\chi := \frac{\chi \chi^H}{\chi^H \chi} = \frac{1}{2}(\mathbf{I}_2 + \mathbf{s} \cdot \boldsymbol{\sigma}). \quad (8.3)$$

For $\chi' = \mathbf{R}\chi$,

$$\rho_{\chi'} = \mathbf{R} \rho_\chi \mathbf{R}^H, \quad \text{tr}(\rho_{\chi'} \text{mat}(Z)) = \text{tr}(\rho_\chi \mathbf{R}^H \text{mat}(Z) \mathbf{R}). \quad (8.4)$$

Thus rotating the state is dual to conjugating the observable. For a spinor $\psi_R := R\psi$, expectation values transform as

$$\psi_R^H = \psi^H R^H, \quad \langle Z \rangle_{\psi_R} = \frac{\text{sc}(\psi^H R^H Z R \psi)}{\text{sc}(\psi^H \psi)}, \quad R^H = R^{-1}. \quad (8.5)$$

For $Z = \mathbf{u} \cdot \boldsymbol{\sigma}$, this gives $Z' = R^H Z R = \mathbf{u}' \cdot \boldsymbol{\sigma}$.

8.2 Boosts

Let rapidity θ satisfy $\beta = \tanh \theta$ and $\gamma = \cosh \theta$. A boost along $\mathbf{u}$ is

$$L = \exp\left(-\frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}\right), \quad L^H = L, \quad L^{-1} = L^*. \quad (8.6)$$

For $\psi_L := L\psi$, the boost generator is Hermitian, so

$$\psi_L^H = \psi^H L, \quad \langle Z \rangle_{\psi_L} = \frac{\text{sc}(\psi^H L Z L \psi)}{\text{sc}(\psi^H L^2 \psi)}. \quad (8.7)$$

The similarity $L^{-1} Z L$ is a different operation: it preserves $\text{sc}(Z)$ and is not the spacetime or spinor-bilinear boost used here. The spacetime paravector therefore transforms by the Hermitian congruence $X' = L X L$, which yields the components $\mathbf{r}_\parallel = (\mathbf{r} \cdot \mathbf{u})\mathbf{u}$, $\mathbf{r}_\perp = \mathbf{r} - \mathbf{r}_\parallel$, $\mathbf{r}'_\parallel = (\mathbf{r}' \cdot \mathbf{u})\mathbf{u}$, and $\mathbf{r}'_\perp = \mathbf{r}' - \mathbf{r}'_\parallel$:

$$t' = \gamma(t - \beta \mathbf{u} \cdot \mathbf{r}), \quad (8.8)$$

$$\mathbf{r}'_\parallel = \gamma(\mathbf{r}_\parallel - \beta t \mathbf{u}), \quad \mathbf{r}'_\perp = \mathbf{r}_\perp. \quad (8.9)$$

It preserves $\det(X) = t^2 - \mathbf{r}^2$. The electromagnetic field instead obeys $F' = LFL^*$ for $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$. Define $\mathbf{E}_{\parallel} = (\mathbf{E} \cdot \mathbf{u})\mathbf{u}$, $\mathbf{E}_{\perp} = \mathbf{E} - \mathbf{E}_{\parallel}$ and likewise $\mathbf{E}'_{\parallel} = (\mathbf{E}' \cdot \mathbf{u})\mathbf{u}$, $\mathbf{E}'_{\perp} = \mathbf{E}' - \mathbf{E}'_{\parallel}$; define the four corresponding components of $\mathbf{B}$ and $\mathbf{B}'$ by the same formulas. Then

$$\mathbf{E}'_{\parallel} = \mathbf{E}_{\parallel}, \quad \mathbf{E}'_{\perp} = \gamma(\mathbf{E}_{\perp} + \beta \mathbf{u} \times \mathbf{B}), \quad (8.10)$$

$$\mathbf{B}'_{\parallel} = \mathbf{B}_{\parallel}, \quad \mathbf{B}'_{\perp} = \gamma(\mathbf{B}_{\perp} - \beta \mathbf{u} \times \mathbf{E}). \quad (8.11)$$

8.3 Congruences, transformed bases, and non-inertial terms

Spinor bilinears use the Hermitian congruence $Y = T^H X T$. With $d\Theta = T^{-1} dT$, direct differentiation gives the ancillary identity

$$dY = T^H(dX)T + (d\Theta)^H Y + Y d\Theta. \quad (8.12)$$

For a rotation, $T^H = T^{-1}$ and $(d\Theta)^H = -d\Theta$, so this reduces to a commutator. For a Hermitian boost it does not: the two connection terms form the congruence appropriate to the spinor bilinear.

Transformed algebra bases use the distinct similarity transformation. For any invertible T , let $X' = T^{-1} X T$ and $d\Theta = T^{-1} dT$. Differentiation gives the useful connection terms

$$dX' = T^{-1}(dX)T + [X', d\Theta], \quad (8.13)$$

$$d^2 X' = T^{-1}(d^2 X)T + 2[dX', d\Theta] + [X', d^2 \Theta] + [d\Theta, [X', d\Theta]]. \quad (8.14)$$

In time-parametrized form, $\Omega = T^{-1} dT/dt$ gives

$$\frac{dX'}{dt} = T^{-1} \frac{dX}{dt} T + [X', \Omega], \quad (8.15)$$

$$\frac{d^2 X'}{dt^2} = T^{-1} \frac{d^2 X}{dt^2} T + 2 \left[\frac{dX'}{dt}, \Omega \right] + \left[X', \frac{d\Omega}{dt} \right] + [\Omega, [X', \Omega]]. \quad (8.16)$$

The transformed generators $\sigma'_n = T^{-1} \sigma_n T$ obey the same algebra, including $(\sigma'_n)^2 = 1$ and $\sigma'_1 \sigma'_2 \sigma'_3 = i$, and

$$d\sigma'_n = [\sigma'_n, T^{-1} dT]. \quad (8.17)$$

If $T = f + \mathbf{g} \cdot \boldsymbol{\sigma}$, then

$$T^{-1} = \frac{f - \mathbf{g} \cdot \boldsymbol{\sigma}}{f^2 - \mathbf{g}^2}, \quad (8.18)$$

$$T^{-1} dT = \frac{f df - \mathbf{g} \cdot d\mathbf{g}}{f^2 - \mathbf{g}^2} + \frac{f d\mathbf{g} - df \mathbf{g} - i\mathbf{g} \times d\mathbf{g}}{f^2 - \mathbf{g}^2} \cdot \boldsymbol{\sigma}. \quad (8.19)$$

Write the vector part of Ω in the primed basis as

$$\text{vec}(\Omega) = \frac{1}{2}(\mathbf{k} + i\boldsymbol{\omega}), \quad k_n + i\omega_n = \text{sc}(2(dT/dt)T^{-1}\sigma_n). \quad (8.20)$$

Then

$$\frac{d\boldsymbol{\sigma}'}{dt} = (-\boldsymbol{\omega} + i\mathbf{k}) \times \boldsymbol{\sigma}', \quad (8.21)$$

and, for $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}'$,

$$\frac{dZ}{dt} = \frac{dS}{dt} + \left(\frac{d\mathbf{V}}{dt} - \mathbf{V} \times \boldsymbol{\omega} + i\mathbf{V} \times \mathbf{k} \right) \cdot \boldsymbol{\sigma}'. \quad (8.22)$$

For $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}'$ these are the rotational and boost-like inertial terms in the measured velocity. When $T^{-1} = \text{adj}(T)$, the transformed basis satisfies $\text{adj}(\sigma'_n) = -\sigma'_n$ and $\det(Z) = S^2 - \mathbf{V}^2$ for $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}'$.

Collinear boosts add rapidities and therefore obey

$$\beta' = \frac{\beta_1 + \beta_2}{1 + \beta_1 \beta_2}, \quad \gamma' = \gamma_1 \gamma_2 (1 + \beta_1 \beta_2). \quad (8.23)$$

Products reverse under inversion:

$$(LR)^{-1} = R^H L^*, \quad (RL)^{-1} = L^* R^H, \quad (R_1 R_2)^{-1} = R_2^H R_1^H, \quad (L_1 L_2)^{-1} = L_2^* L_1^*. \quad (8.24)$$

9 Spectral functions and quaternions

9.1 Functions of a unit vector

With the algebra projectors $\Pi_{\pm} := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})$, any function defined by a power series satisfies

$$f(\mathbf{u} \cdot \boldsymbol{\sigma}) = f(1)\Pi_+ + f(-1)\Pi_- = \frac{1}{2}(f(1) + f(-1)) + \frac{1}{2}(f(1) - f(-1))\mathbf{u} \cdot \boldsymbol{\sigma}. \quad (9.1)$$

Consequently,

$$\sqrt{\mathbf{u} \cdot \boldsymbol{\sigma}} = \pm \Pi_+ \pm i \Pi_-, \quad (9.2)$$

$$\log(\mathbf{u} \cdot \boldsymbol{\sigma}) = 2i\pi(n\Pi_+ + (\frac{1}{2} + m)\Pi_-), \quad (9.3)$$

$$(z + \mathbf{u} \cdot \boldsymbol{\sigma})^{-1} = \frac{z - \mathbf{u} \cdot \boldsymbol{\sigma}}{z^2 - 1}, \quad (9.4)$$

$$e^{\theta \mathbf{u} \cdot \boldsymbol{\sigma}} = \cosh \theta + \sinh \theta \mathbf{u} \cdot \boldsymbol{\sigma}, \quad (9.5)$$

$$e^{i\theta \mathbf{u} \cdot \boldsymbol{\sigma}} = \cos \theta + i \sin \theta \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (9.6)$$

For $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$, put $\lambda_{\pm} = t \pm \|\mathbf{r}\|$ and $\mathbf{u} = \mathbf{r}/\|\mathbf{r}\|$. Then

$$X = \lambda_+ \Pi_+ + \lambda_- \Pi_-, \quad f(X) = f(\lambda_+) \Pi_+ + f(\lambda_-) \Pi_-. \quad (9.7)$$

This includes

$$X^n = \frac{1}{2}(\lambda_+^n + \lambda_-^n) + \frac{1}{2}(\lambda_+^n - \lambda_-^n)\mathbf{u} \cdot \boldsymbol{\sigma}, \quad (9.8)$$

$$X^{-1} = \lambda_+^{-1} \Pi_+ + \lambda_-^{-1} \Pi_-, \quad (9.9)$$

$$\sqrt{X} = \pm \sqrt{\lambda_+} \Pi_+ \pm \sqrt{\lambda_-} \Pi_-, \quad (9.10)$$

$$\log X = \log \lambda_+ \Pi_+ + \log \lambda_- \Pi_-. \quad (9.11)$$

For two directions,

$$\Pi_+ \Pi'_+ = \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(\mathbf{u} + \mathbf{u}' + i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (9.12)$$

$$\Pi_+ \Pi'_- = \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(\mathbf{u} - \mathbf{u}' - i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (9.13)$$

$$\Pi_- \Pi'_+ = \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(-\mathbf{u} + \mathbf{u}' - i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (9.14)$$

$$\Pi_- \Pi'_- = \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(-\mathbf{u} - \mathbf{u}' + i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}. \quad (9.15)$$

One useful projector evaluation is

$$\frac{1 - \mathbf{r} \cdot \boldsymbol{\sigma}}{1 + \mathbf{r} \cdot \boldsymbol{\sigma}} = e^{-2 \text{atanh}(\|\mathbf{r}\|)} \Pi_+ + e^{2 \text{atanh}(\|\mathbf{r}\|)} \Pi_-. \quad (9.16)$$

9.2 Quaternion subalgebra

Define

$$q_1 = -\sigma_2\sigma_3, \quad q_2 = -\sigma_3\sigma_1, \quad q_3 = -\sigma_1\sigma_2, \quad \mathbf{q} = -\mathbf{i}\boldsymbol{\sigma}. \quad (9.17)$$

Then $q_1^2 = q_2^2 = q_3^2 = q_1q_2q_3 = -1$. A quaternion $Q = a + \mathbf{b} \cdot \mathbf{q}$ satisfies $Q^H = a - \mathbf{b} \cdot \mathbf{q}$ and $QQ^H = a^2 + \mathbf{b}^2$; the rotation action is $\mathbf{r}' \cdot \mathbf{q} = Q(\mathbf{r} \cdot \mathbf{q})Q^H$.

10 Generalized spacetime and metric form

Let

$$X = f(t, \mathbf{r}) + \mathbf{g}(t, \mathbf{r}) \cdot \boldsymbol{\sigma}, \quad dX = df + d\mathbf{g} \cdot \boldsymbol{\sigma}. \quad (10.1)$$

Writing $dX = A dt + \mathbf{B} \cdot d\mathbf{r}$ gives

$$A = \partial_t f + \partial_t \mathbf{g} \cdot \boldsymbol{\sigma}, \quad \mathbf{B} = \partial_{\mathbf{r}} f + \partial_{\mathbf{r}}(\mathbf{g} \cdot \boldsymbol{\sigma}), \quad (10.2)$$

and the interval remains

$$ds^2 = dX dX^* = df^2 - d\mathbf{g}^2. \quad (10.3)$$

Define the geometric velocity and Lorentz factor by

$$\mathbf{v} = \frac{d\mathbf{g}}{df}, \quad \gamma = \frac{df}{ds} = \frac{1}{\sqrt{1 - \mathbf{v}^2}}, \quad U = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (10.4)$$

With geometric acceleration $\mathbf{a} = d\mathbf{v}/ds = \gamma d^2\mathbf{g}/df^2$, the proper-acceleration formula is unchanged. Inertial motion $dU/ds = 0$ implies $\mathbf{a} = 0$, hence $\mathbf{g} = c_1 f + c_2$ and constant geometric velocity. If $f' = df/dt$ and $\mathbf{g}' = d\mathbf{g}/dt$, then

$$\frac{d\mathbf{v}}{dt} = \frac{\mathbf{g}''f' - \mathbf{g}'f''}{(f')^2}, \quad [U, dU/ds] = 2\gamma^3(\mathbf{v} \times \mathbf{a}) \cdot \mathbf{i}\boldsymbol{\sigma}. \quad (10.5)$$

In coordinates $\mathbf{x} = \{t, r_1, r_2, r_3\}^\top$, write

$$ds^2 = d\mathbf{x}^\top \mathbf{Q}(\mathbf{x}) d\mathbf{x}. \quad (10.6)$$

For real symmetric $\mathbf{Q} = \mathbf{N} \mathbf{D} \mathbf{N}^\top$ with eigenvalues λ_n and orthonormal eigenvectors $\mathbf{u}_n$,

$$ds^2 = \sum_n \lambda_n (\mathbf{u}_n \cdot d\mathbf{x})^2, \quad dx'_n = \sqrt{\lambda_n} \mathbf{u}_n \cdot d\mathbf{x}, \quad \mathbf{Q}^{-1} = \mathbf{N} \mathbf{D}^{-1} \mathbf{N}^\top. \quad (10.7)$$

11 Reference summary

This index collects the principal definitions and results in natural units, with signature $(+, -, -, -)$. Topic titles and equation numbers link to their full statements and derivations.

Algebra and vector operations.

$$\begin{aligned} \sigma_n \sigma_m &= \delta_{nm} + \epsilon_{nmk} \mathbf{i} \sigma_k, & [\sigma_n, \sigma_m] &= 2\epsilon_{nmk} \mathbf{i} \sigma_k & (1.5), \\ (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) &= \mathbf{A} \cdot \mathbf{B} + \mathbf{i}(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, & \mathbf{A}^2 &= \mathbf{A} \cdot \mathbf{A}, \\ \|Z\|^2 &= \|S\|^2 + \|\mathbf{V}\|^2, & \|\mathbf{V}\|^2 &= \mathbf{V} \cdot \mathbf{V}^* = \mathbf{V}^* \cdot \mathbf{V} & (4.21). \end{aligned}$$

Involutions and adjugation.

$$\begin{aligned} (ZZ')^* &= Z^* Z'^*, & (ZZ')^H &= Z'^H Z^H, \\ \text{adj}(Z) &= Z^{*H}, & \text{adj}(ZZ') &= Z'^* Z^{*H} \end{aligned} \quad (2.35).$$

Paravectors, extraction, and products. For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma} = (a + bi) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$,

$$\begin{aligned} S &= \text{sc}(Z) = \frac{1}{2}(Z + Z^{*H}), & \mathbf{V} \cdot \boldsymbol{\sigma} &= \frac{1}{2}(Z - Z^{*H}), \\ a &= \text{sc}(\text{re}(Z)), & b &= \text{sc}(\text{im}(Z)), \\ \mathbf{A} &= \text{vec}(\text{re}(Z)), & \mathbf{B} &= \text{vec}(\text{im}(Z)), \\ ZZ' &= (SS' + \mathbf{V} \cdot \mathbf{V}') + (S\mathbf{V}' + S'\mathbf{V} + i\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma}, \\ \det(Z) &= \text{adj}(Z) Z = Z \text{adj}(Z) = S^2 - \mathbf{V}^2, & Z^{-1} &= \frac{\text{adj}(Z)}{\det(Z)}, \\ Z^2 - 2SZ + (S^2 - \mathbf{V}^2) &= 0, & \lambda_{\pm} &= S \pm \sqrt{\mathbf{V}^2}. \end{aligned}$$

See the product (4.15), determinant (4.18), and quadratic identity (4.29).

Pauli-matrix representation.

$$\begin{aligned} \text{mat}(Z) &= \begin{pmatrix} S + V_3 & V_1 - iV_2 \\ V_1 + iV_2 & S - V_3 \end{pmatrix}, & \text{tr}(\mathbf{M}) &= 2S \\ \mathbf{M}^H &= (\mathbf{M}^C)^T = (\mathbf{M}^T)^C, & \mathbf{M}^* &= -\mathbf{J}\mathbf{M}^C\mathbf{J}, \\ \text{mat}(\text{adj}(Z)) &= \text{adj}(\mathbf{M}), & \text{adj}(\mathbf{M}) &= \text{tr}(\mathbf{M})\mathbf{I}_2 - \mathbf{M}. \end{aligned} \quad (2.2),$$

Space-time kinematics and action.

$$\begin{aligned} dX &= dt + d\mathbf{r} \cdot \boldsymbol{\sigma}, & ds^2 &= dX dX^* = dt^2 - d\mathbf{r}^2 & (5.2), \\ U &= \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), & \gamma &= (1 - \mathbf{v}^2)^{-1/2}, \\ mU &= E + \mathbf{P} \cdot \boldsymbol{\sigma}, & E^2 - \mathbf{P}^2 &= m^2 & (5.11), \\ \int L dt &= -m \int ds, & L &= -m/\gamma, \quad H = m\gamma. \end{aligned}$$

Electromagnetism.

$$\begin{aligned} \partial &= \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, & F &= (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, & D &= \rho + \mathbf{J} \cdot \boldsymbol{\sigma}, \\ \partial F &= D^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}, & \square &= \partial^* \partial & (6.2), \\ \partial \Phi &= S + F^*, & \mathbf{E} &= -\partial_t \mathbf{A} - \partial_{\mathbf{r}} V, & \mathbf{B} &= \partial_{\mathbf{r}} \times \mathbf{A}, \\ \frac{1}{2}F^2 &= \frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2) + i\mathbf{E} \cdot \mathbf{B}, & \frac{1}{2}FF^H &= \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ m \frac{dU}{ds} &= q \text{re}(FU), & \frac{d\mathbf{P}}{dt} &= q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) & (6.29). \end{aligned}$$

Wave equations, spin, and energy sectors.

$$\begin{aligned}
\langle \psi' | \psi \rangle &= \text{sc}(\psi'^{\mathbf{H}} \psi) = \frac{1}{2} \text{tr}(\mathbf{M}_{\psi'}^{\mathbf{H}} \mathbf{M}_{\psi}), \\
\langle \psi' | Z | \psi \rangle &= \text{sc}(\psi'^{\mathbf{H}} Z \psi) = \frac{1}{2} \text{tr}(\mathbf{M}_{\psi'}^{\mathbf{H}} \text{mat}(Z) \mathbf{M}_{\psi}), \\
\langle Z \rangle_{\psi} &= \frac{\langle \psi | Z | \psi \rangle}{\langle \psi | \psi \rangle} = \frac{\text{sc}(\psi^{\mathbf{H}} Z \psi)}{\text{sc}(\psi^{\mathbf{H}} \psi)}, \\
(\square + m^2) \Psi &= 0, \\
\mathbf{W}(Z) &= \begin{pmatrix} 0 & Z \\ \text{adj}(Z) & 0 \end{pmatrix}, \\
(\mathbf{W}(\text{i}\partial) - m\mathbf{I}) \Psi &= 0, \\
(\mathbf{W}(\text{i}\partial - q\Phi^*) - m\mathbf{I}) \Psi &= 0, \\
\text{i}\partial_t \Psi &= \left(\frac{(-\text{i}\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV - \frac{q}{2m} \mathbf{B} \cdot \boldsymbol{\sigma} \right) \Psi, \\
\Pi_{\pm}(\mathbf{u}) &= \frac{1}{2}(\mathbf{I}_2 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\
p_{\pm}(\mathbf{u}) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \mathbf{s}), \\
\Lambda_{\pm} &= \frac{1}{2}(\mathbf{I} \pm \mathbf{W}(U)), \quad \Pi_{j,4} = \text{diag}(\Pi_j, \Pi_j), \\
\Psi &= \sum_{i=\pm} \sum_{j=\pm} \Psi_{ij}, \quad \Psi_{ij} = \Lambda_i \Pi_{j,4} \Psi.
\end{aligned}$$

See the paravector inner product (7.1), minimal-ideal spinor (7.13), Klein–Gordon equation (7.15), Dirac equations (7.20) and (7.22), Pauli equation (7.23), and spin probability (7.51).

Rotations, boosts, and transformed bases.

$$\begin{aligned}
R &= e^{\phi \mathbf{u} \cdot \mathbf{i}\boldsymbol{\sigma}/2}, \quad R^{\mathbf{H}} = R^{-1}, \quad Z' = R^{\mathbf{H}} Z R, \\
L &= e^{-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2}, \quad L^{\mathbf{H}} = L, \quad X' = L X L, \\
\langle Z \rangle_{\psi_T} &= \frac{\text{sc}(\psi^{\mathbf{H}} T^{\mathbf{H}} Z T \psi)}{\text{sc}(\psi^{\mathbf{H}} T^{\mathbf{H}} T \psi)}, \\
\text{d}(T^{\mathbf{H}} X T) &= T^{\mathbf{H}}(\text{d}X)T + (\text{d}\Theta)^{\mathbf{H}} T^{\mathbf{H}} X T + T^{\mathbf{H}} X T \text{d}\Theta, \\
X' &= T^{-1} X T, \\
\text{d}X' &= T^{-1}(\text{d}X)T + [X', T^{-1} \text{d}T].
\end{aligned}$$

See the bilinear rule (7.4) and congruence derivative (8.12).

Spectral functions and quaternions.

$$\begin{aligned}
f(\mathbf{u} \cdot \boldsymbol{\sigma}) &= f(1)\Pi_+ + f(-1)\Pi_-, \\
X &= \lambda_+ \Pi_+ + \lambda_- \Pi_-, \quad f(X) = f(\lambda_+)\Pi_+ + f(\lambda_-)\Pi_-, \\
\mathbf{q} &= -\text{i}\boldsymbol{\sigma}, \quad Q^{\mathbf{H}} = a - \mathbf{b} \cdot \mathbf{q}, \quad QQ^{\mathbf{H}} = a^2 + \mathbf{b}^2.
\end{aligned}$$

Generalized spacetime and metric form.

$$\begin{aligned} X &= f + \boldsymbol{g} \cdot \boldsymbol{\sigma}, & \mathrm{d}s^2 &= \mathrm{d}f^2 - \mathrm{d}\boldsymbol{g}^2, \\ \boldsymbol{v} &= \mathrm{d}\boldsymbol{g}/\mathrm{d}f, & U &= \gamma(1 + \boldsymbol{v} \cdot \boldsymbol{\sigma}), \\ \mathrm{d}s^2 &= \mathrm{d}\boldsymbol{x}^\top \boldsymbol{Q}(\boldsymbol{x}) \mathrm{d}\boldsymbol{x}, & \boldsymbol{Q} &= \boldsymbol{N} \boldsymbol{D} \boldsymbol{N}^\top. \end{aligned}$$

Conventions and cautions

- The same symbol $\mathbf{i} = \sigma_1 \sigma_2 \sigma_3$ is both the algebra's pseudoscalar and, in the Pauli representation, the ordinary complex unit.
- Sigma conjugation $(\cdot)^*$, reversion $(\cdot)^H$, and their composition are distinct. In the Pauli representation, reversion corresponds to Hermitian adjunction.
- $\det(Z)$ is an indefinite quadratic form and may be complex for complex Z .
- Differential operators act on everything to their right unless parentheses state otherwise. This matters in products involving fields.

*Editorial notation record.*¹

¹Compared with `tmp.txt`, the following presentation and style changes were made. Vectors and matrices are bold, and only vectors and matrices are bold; in particular, the electromagnetic fields formerly written as calligraphic or italic E and B are the upright bold vectors $\boldsymbol{E}$ and $\boldsymbol{B}$, while the scalar energy remains the unbold symbol E . The action is written as $\int L \mathrm{d}t$, avoiding conflict with the scalar part S ; the Lagrangian L and Hamiltonian H use the regular math font rather than decorative alphabets. Blackboard bold is retained only for the number sets $\mathbb{R}$ and $\mathbb{C}$. The Dirac block function is the upright bold $\boldsymbol{W}(\cdot)$. A vector square is explicitly defined by $\boldsymbol{A}^2 := \boldsymbol{A} \cdot \boldsymbol{A}$. Curly braces are reserved for explicit vector-entry lists, and commutators alone use square brackets. Parentheses delimit matrices, functions, algebraic grouping, and order-sensitive operator products; redundant grouping parentheses were removed. The extraction functions $\mathrm{sc}(\cdot)$, $\mathrm{vec}(\cdot)$, $\mathrm{re}(\cdot)$, and $\mathrm{im}(\cdot)$ replace `scalar`, `vector`, and capitalized variants and are defined before use; the unused expectation abbreviation `Exp` was removed. The original metric notation $\langle\langle Z \rangle\rangle^2$ and every determinant bar were replaced by the unambiguous function notation $\det(Z)$. Double bars are reserved for squared scalar, vector, and coefficient norms, with $\|Z\|^2 = \|S\|^2 + \|\boldsymbol{V}\|^2$; absolute-value bars are not used. Entrywise complex-conjugation notation is confined to the matrix section. Sigma conjugation retains the splat `*`; reversal, originally ^{R} and elsewhere a dagger, is written ^{H} , because it reverses the multiplied sigma order and becomes Hermitian adjunction in the matrix representation. The abbreviation $\mathrm{adj}(Z) := Z^{*H}$ is defined early and used where it shortens formulas. The matrix section uses matching upright sans-serif capitals ^{$\top$} , ^{$\mathbb{C}$} , and ^{H} for transpose, entrywise complex conjugation, and Hermitian transpose, and distinguishes these from sigma conjugation, including which operations reverse product order. Plus/minus labels are subscripts ($\pm$, $+$, $-$), not superscripts or parenthesized labels. Parallel and perpendicular components use $\parallel$ and $\perp$ rather than `par` and `prp`, with their projection formulas restated locally at each use. Spatial derivatives use ∂_r rather than ∇ , and no overbars are used. Literal or malformed source spacing and fraction commands such as `quad` and `frac` were converted to properly formatted L^AT_EX. Long product expansions were consolidated while retaining their distinct components and results, and expressions otherwise preserve the compact ordering of the source, for example $\mathrm{sc}(\Phi^* U)$ rather than adding an extra pair of parentheses.